Hamiltonians of mean force

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We study the operator representability problem for the Hamiltonian of mean force (HMF) in Gaussian linearly coupled environments beyond the commuting limit. Building on the quenched-density construction, we derive an exact imaginary-time reformulation in which bath statistics enter as scalar kernel moments while operator growth is generated by the adjoint chain $\{\mathrm{ad}^n_{H_Q}(f)\}_{n\geq 0}$. This yields a structural criterion: a finite closed-form HMF exists only when the adjoint-generated operator algebra closes inside the target ansatz. We then instantiate the criterion in a Pauli-closed qubit model and obtain an explicit closed-form mean-force Hamiltonian with temperature- and bath-dependence carried by mode-resolved kernel coefficients. The manuscript is framed as the noncommuting Gaussian follow-up to the commuting benchmark analysis of arXiv:2602.13146 and sets up the non-Gaussian extension program.

INTRODUCTION

This manuscript is a direct follow-up to Ref. [1] (arXiv:2602.13146). That paper established the quenched-density representation and delivered an exact benchmark in the commuting Gaussian sector. The present paper starts from that benchmark and addresses the next step: how the Hamiltonian of mean force is organised once the coupling operator does *not* commute with the bare system Hamiltonian.

In closed quantum statistical mechanics, equilibrium is generated by a Hamiltonian: $\rho \propto e^{-\beta H}$. For an open system with finite coupling, the operationally defined equilibrium state of the subsystem is the reduced state of the global Gibbs ensemble,

$$\bar{\rho}_Q(\beta) = \mathrm{Tr}_X\, e^{-\beta H_{\mathrm{tot}}}, \tag{1}$$

$$e^{-\beta H_{\mathrm{MF}}(\beta)} \propto \frac{\mathrm{Tr}_X e^{-\beta H_{\mathrm{tot}}}}{Z_X(\beta)}, \quad Z_X(\beta) = \mathrm{Tr}_X e^{-\beta H_X}. \tag{2}$$

This object is generally not $e^{-\beta H_Q}$ for the bare system Hamiltonian H_Q. The resulting representational question is precise: what operator, if any, plays the role of an equilibrium generator for the subsystem once the coupling is non-negligible? The Hamiltonian of mean force (HMF) answers this by construction and is the standard starting point in strong-coupling thermodynamics and the mean-force Gibbs-state program [2–7].

Why care about a mean-force Hamiltonian at all? First, it provides the exact reduced equilibrium object that underlies strong-coupling thermodynamic identities, including free-energy and work relations that remain valid beyond weak coupling[2, 5, 8–11]. Second, it furnishes a consistent equilibrium initialization for open-system dynamics when correlations with the bath are unavoidable, a point emphasized in the literature on correlated initial states and reduced dynamics [4, 12]. Related subensemble approaches explicitly treat the subsystem thermodynamics as inherited from a global canonical ensemble, making the effective reduced generator central to the formalism[13]. In short, the HMF is not an optional reinterpretation; it is the exact operator that encodes the reduced equilibrium state whenever system–bath coupling is finite.

Historically, open-system theory prioritized weak-coupling and Markovian regimes, where reduced equilibrium can often be approximated by a Gibbs state of a renormalized H_Q. At finite coupling the reduced state inherits explicit temperature dependence and interaction-induced operator content that is not captured by a simple renormalization. Coupling-dependent thermodynamic response features in quantum Brownian motion and related models highlight this complexity[14, 15]. The strong-coupling literature consequently treats the mean-force Gibbs state as a distinct equilibrium object, with operational ramifications for heat and energy definitions[9, 16–18]. The HMF can encode interaction-induced operator content absent from H_Q [14, 15, 19], making explicit representation of this content a primary goal.

The HMF literature itself is broad but structured. Canonical definitions and thermodynamic identities are developed in strong-coupling thermodynamics and fluctuation-relation work[2, 3, 5, 8]. A comprehensive review consolidates the "static" mean-force Gibbs perspective with the "dynamical" return-to-equilibrium perspective in open quantum systems [4]. Operational questions such as measurability and thermodynamic consistency at strong coupling have also been pursued[9, 17, 18].

Exact or controlled evaluations exist in special cases. For commuting (QND) interactions, the operator algebra closes trivially and the HMF can be written explicitly[20]. Quadratic/Gaussian models (e.g., damped harmonic oscillators) are solvable because Gaussianity is preserved, leading to closed operator forms[21–23]. Finite-dimensional closures such as spin-boson or single-qubit models provide additional controlled benchmarks[24]. Beyond these cases, the mean-force Gibbs state is often accessed through systematic limits: Cresser and Anders derive weak- and ultrastrong-coupling expressions and

show that, in the ultrastrong limit, the mean-force Gibbs state becomes diagonal in the interaction basis rather than the system Hamiltonian basis[25, 26]. Recent work generalizes the HMF framework to finite baths by introducing a pair of quantum Hamiltonians of mean force that incorporate bath feedback[27], and structural studies further analyze the operator content and potential renormalization of the HMF in extended settings[19, 28].

Outside solvable models, most approaches are perturbative or numerical. Weak-coupling expansions yield controlled but limited series [25, 29]. Reaction-coordinate mappings[30–33] and polaron transforms provide approximate local generators applicable at intermediate coupling. Imaginary-time path-integral methods, stochastic representations, and hierarchical-equations techniques can compute $\bar{\rho}_Q(\beta)$ directly but typically do not yield compact operator forms for $H_{\rm MF}$[34–39]. Stochastic Liouville and partition-free approaches give complementary numerical access without yielding closed-form HMFs [40, 41]. These methods are indispensable for quantitative predictions but leave open the representational question: when does a local or otherwise restricted operator form exist?

The influence-functional formalism is a natural language for this problem. For Gaussian baths with linear coupling it provides an exact route to integrating out bath degrees of freedom[21, 22, 42]. In equilibrium it becomes a Euclidean (imaginary-time) influence functional, usually bilocal in τ, and admits a Hubbard–Stratonovich rewriting as a quenched Gaussian-field average[43–45]. Related path-integral derivations of quantum Langevin dynamics make explicit the noise and dissipation structure inherited from the bath[46, 47]. This formulation also clarifies the terminology: "nonlocal" initially refers to imaginary-time nonlocality of the kernel, whereas the HMF locality question concerns the operator structure on the system Hilbert space.

The existence of $H_{\rm MF}$ is therefore not the issue—it is defined by a logarithm of a traced exponential. The real obstruction is representability: when does $H_{\rm MF}$ admit a closed-form expression within a restricted operator family (few-body, spatially local, or a given algebra)? Much of the literature either (i) solves special models, (ii) expands in controlled limits, or (iii) computes $\bar{\rho}_Q(\beta)$ numerically and analyzes its properties, but a general structural criterion is still lacking [25, 27, 28].

Relative to Ref. [1], the present manuscript has a narrower but deeper scope. We retain the exact Gaussian influence-functional and quenched-field identities, then use adjoint-action and Magnus/BCH organisation [48–50] to study representability in the noncommuting sector. The central result is an explicit closure criterion: a finite closed-form HMF exists only when the operator algebra generated by repeated adjoint action of H_Q on the coupling operators closes inside the target ansatz. We then instantiate this criterion in a Pauli-closed qubit example, where all bath dependence is reduced to scalar kernel moments and the mean-force Hamiltonian is written in exact closed form. Moreover, it is not often one has the opportunity to say something new about a qubit.

QUENCHED REPRESENTATION AND INFLUENCE FUNCTIONAL

We begin by recapitulating the quenched representation introduced in Ref. [1]. To proceed directly from Sec. , we explicit the composite model. We denote the bare system Hamiltonian by H_Q and write

$$H_{\rm tot} = H_Q + H_X + H_{\rm int}, \tag{3}$$

where H_X is the bath Hamiltonian. We assume a factorizable interaction

$$H_{\rm int} = f \otimes B, \tag{4}$$

where f acts on the system and B is a bath operator. We can define the reduced equilibrium operator (up to normalisation) by $\bar{\rho}_Q(\beta) \equiv {\rm Tr}_X e^{-\beta H_{\rm tot}}$. As shown in Ref. [1], this can be represented as a *quenched density*. This is an average over a stochastic propagator, given by:

$$\bar{\rho}_Q(\beta) = \mathbb{E}_\xi\big[U_\xi(\beta)\big], \tag{5}$$

$$U_\xi(\beta) \equiv \mathcal{T}_\tau \exp\left[-\int_0^\beta d\tau\,\big(H_Q + \xi(\tau) f\big)\right], \tag{6}$$

where $\xi(\tau)$ is a stochastic process whose statistics encode the bath correlations, and $\mathcal{T}_\tau$ denotes time-ordering in imaginary time. Regardless of the precise form of bath and coupling, the reduced density must always be describable in this form.

This result can be understood intuitively by first observing that since the environment influence can only enter through the system coupling f, its influence can be captured by attaching a τ-dependent driving field to f [51, 52]. If this were a single deterministic field however, it would correspond to the bath exerting the *same* back-action history for every microscopic bath configuration. But ${\rm Tr}_X$ averages over many bath microstates in the thermal ensemble, and hence over many back-action histories. In this sense the partial trace is necessarily an average over histories in imaginary time, and the quenched representation simply makes this averaging explicit. For each realisation $\xi(\tau)$ the system evolves under an imaginary-time Hamiltonian $H_Q + \xi(\tau) f$; the bath is then recovered by averaging over $\xi(\tau)$ with a law chosen to reproduce the bath-induced correlations. In this sense $\xi(\tau)$ is not a physical external control field but an efficient parametrisation of the bath history $B(\tau)$ as seen through the coupling channel.

We may understand what formal properties are demanded of ξ by considering the influence functional it is

required to match. Working in imaginary time, introduce the bath interaction picture:

$$B(\tau) \equiv e^{\tau H_X} B e^{-\tau H_X}, \qquad \tau \in [0, \beta], \tag{7}$$

and define the bath thermal state $\rho_X \equiv e^{-\beta H_X}/Z_X$. For an arbitrary c-number source $j(\tau)$ coupled linearly to $B(\tau)$, the bath generates a (time-ordered) functional

$$\mathcal{Z}_X[j] \equiv \mathrm{Tr}_X \left[\mathcal{T}_\tau \exp\left(-\int_0^\beta d\tau\, j(\tau)\, B(\tau) \right) \rho_X \right]. \tag{8}$$

Here $j(\tau)$ is introduced as an external c-number source used to generate ordered bath correlators by functional differentiation. More generally, $j(\tau)$ may be any object commuting with the bath algebra (e.g. a system operator tensored with $\mathbb{I}_X$). In the influence-functional derivation it is ultimately supplied by the system history, which becomes a c-number function in the path-integral representation. In the influence-functional approach, tracing out the bath produces precisely such a functional, evaluated on the system history through the coupling channel.

From this, the bath contribution to the effective Euclidean action can be written as [21, 22, 42]

$$\mathcal{F}[f] \equiv \mathcal{Z}_X[f], \qquad \Phi[f] \equiv \log \mathcal{F}[f]. \tag{9}$$

A key structural fact is that $\Phi[f] = \log \mathcal{F}[f]$ is the *cumulant generating functional* of the bath operator $B(\tau)$ with respect to the thermal state. Concretely, the generalised (time-ordered) cumulant theorem implies the connected expansion [53, 54]

$$\begin{aligned} \Phi[f] = \sum_{n\geq 1} \frac{(-1)^n}{n!} \int_0^\beta d\tau_1 \cdots d\tau_n \\ \times K^{(n)}(\tau_1, \ldots, \tau_n)\, f(\tau_1) \cdots f(\tau_n), \end{aligned} \tag{10}$$

where the kernels $K^{(n)}$ are the *connected* (cumulant) bath correlators, defined via [55]

$$K^{(n)}(\tau_1, \ldots, \tau_n) \equiv \left\langle \mathcal{T}_\tau B(\tau_1) \cdots B(\tau_n) \right\rangle_c. \tag{11}$$

The explicit connection back to Eq. (8) is given via its functional differentiation:

$$\left\langle \mathcal{T}_\tau B(\tau_1) \cdots B(\tau_n) \right\rangle_c = (-1)^n \left. \frac{\delta^n \log \mathcal{Z}_X[j]}{\delta j(\tau_1) \cdots \delta j(\tau_n)} \right|_{j=0}. \tag{12}$$

The bath influence is then completely characterised by the hierarchy $\{K^{(n)}\}_{n\geq 1}$.

To connect the influence functional back to the quenched density, we use the fact that the bath influence depends on the system history only through the linear functional $\int_0^\beta d\tau\, f(\tau) B(\tau)$. One may therefore represent $\mathcal{F}[f]$ as the (generalised) characteristic functional of an auxiliary field $\xi(\tau)$ [43–45]:

$$\mathcal{F}[f] = \mathbb{E}_\xi \left[\exp\left(-\int_0^\beta d\tau\, \xi(\tau)\, f(\tau) \right) \right], \tag{13}$$

where $\mathbb{E}_\xi[\cdot]$ denotes averaging with respect to a (possibly complex) measure on ξ-histories chosen such that Eq. (13) holds.

Since $\xi(\tau)$ is a commuting c-number field, its n-point moments are symmetric under permutations of the time arguments. The influence kernels $K^{(n)}$ fix the *cumulants* of ξ via

$$\left\langle \xi(\tau_1) \cdots \xi(\tau_n) \right\rangle_c = K^{(n)}(\tau_1, \ldots, \tau_n). \tag{14}$$

Consequently, the ordinary correlation functions of the noise are obtained from $\{K^{(m)}\}$ by the standard moment-cumulant relations:

$$\left\langle \xi(\tau_1) \cdots \xi(\tau_n) \right\rangle = \sum_{\pi \in \mathcal{P}_n} \prod_{C \in \pi} K^{(|C|)}\big(\{\tau_i\}_{i\in C}\big), \tag{15}$$

where $\mathcal{P}_n$ denotes the set of all partitions of the index set $\{1, \ldots, n\}$, and C are the disjoint blocks of a given partition $\pi \in \mathcal{P}_n$. After shifting the mean so that $K^{(1)}(\tau) = \langle \xi(\tau) \rangle = 0$, one has (for example)

$$\left\langle \xi(\tau_1)\xi(\tau_2) \right\rangle = K^{(2)}(\tau_1, \tau_2), \tag{16}$$

$$\left\langle \xi(\tau_1)\xi(\tau_2)\xi(\tau_3) \right\rangle = K^{(3)}(\tau_1, \tau_2, \tau_3), \tag{17}$$

$$\begin{aligned} &\left\langle \xi(\tau_1)\xi(\tau_2)\xi(\tau_3)\xi(\tau_4) \right\rangle \\ &\quad = K^{(4)}(\tau_1, \tau_2, \tau_3, \tau_4) \\ &\qquad + K^{(2)}(\tau_1, \tau_2) K^{(2)}(\tau_3, \tau_4) \\ &\qquad + K^{(2)}(\tau_1, \tau_3) K^{(2)}(\tau_2, \tau_4) \\ &\qquad + K^{(2)}(\tau_1, \tau_4) K^{(2)}(\tau_2, \tau_3). \end{aligned} \tag{18}$$

QUENCHED DENSITY AND THE HAMILTONIAN OF MEAN FORCE

The Hamiltonian of mean force is, by definition, the operator whose Gibbs form reproduces the system's reduced equilibrium state. Equivalently, it is the *operator logarithm* of the unnormalised reduced equilibrium operator

$$\bar{\rho}_Q(\beta) \equiv \mathrm{Tr}_X e^{-\beta H_{\mathrm{tot}}}, \qquad H_{\mathrm{MF}}(\beta) \;\equiv\; -\frac{1}{\beta} \log \bar{\rho}_Q(\beta), \tag{19}$$

defined up to an additive multiple of the identity (fixed only when normalising the state).

The quenched representation in Eq. (5) supplies an exact stochastic parametrisation of the unnormalised mean-force Gibbs operator $\bar{\rho}_Q(\beta)$ by making the bath trace an explicit average over imaginary-time back-action histories. Combining Eq. (5) with Eq. (19) yields

$$H_{\mathrm{MF}}(\beta) = -\frac{1}{\beta} \log \mathbb{E}_\xi[U_\xi(\beta)]\,. \tag{20}$$

Thus, constructing the mean-force Hamiltonian reduces to evaluating a stochastic average and then compressing

the result via an operator logarithm. The conditions for being able to perform this average exactly is the focus of the present work.

To make the handling of this problem more concrete, we shall specialise the environment to the Caldeira-Leggett model, where the bath is a collection of harmonic oscillators ($H_X = \sum_k \omega_k b_k^\dagger b_k$) and the coupling is linear in bath coordinates ($B = \sum_k c_k x_k$). None of the results that follow are essentially dependent on this choice, and a generalisation to anharmonic environments is (relatively) straightforward. In the interests of comprehensibility however, we restrict our scope to quadratic environments. In this setting, the auxiliary field $\xi(\tau)$ becomes a stationary zero-mean Gaussian process completely characterized by its covariance

$$\mathbb{E}_\xi[\xi(\tau)\xi(\tau')] = K(\tau - \tau'). \tag{21}$$

The kernel $K(\tau)$ is determined by the bath spectral density $J(\omega) = \frac{\pi}{2}\sum_k \frac{c_k^2}{m_k\omega_k}\delta(\omega - \omega_k)$ via the relation [56]

$$K(\tau) = \frac{1}{\pi}\int_0^\infty d\omega\, J(\omega)\,\frac{\cosh\big(\omega(\beta/2 - |\tau|)\big)}{\sinh(\beta\omega/2)}. \tag{22}$$

A key property of this kernel is its integrated strength. Integrating Eq. (22) yields

$$\begin{aligned}\int_0^\beta d\tau\, K(\tau) &= \frac{1}{\pi}\int_0^\infty d\omega\, J(\omega)\int_0^\beta d\tau\,\frac{\cosh\big(\omega(\beta/2 - |\tau|)\big)}{\sinh(\beta\omega/2)}\\ &= \frac{1}{\pi}\int_0^\infty d\omega\, J(\omega)\,\frac{2}{\omega}\\ &= 2\lambda,\end{aligned} \tag{23}$$

where λ is the explicit reorganisation energy. Consequently, the total variance of the integrated noise field $\Xi = \int_0^\beta d\tau\,\xi(\tau)$ grows linearly with inverse temperature:

$$\mathbb{E}_\xi[\Xi^2] = \int_0^\beta d\tau \int_0^\beta d\tau'\, K(\tau - \tau') = 2\beta\lambda. \tag{24}$$

In the case that the system and its coupling commute, $[H_Q, f] = 0$ and f is τ-independent in imaginary time. In this instance time ordering drops out, and the average is given by [1]:

$$\bar{\rho}_Q(\beta) = \exp\left[-\beta\left(H_Q - \frac{\kappa_0(\beta)}{2}f^2\right)\right], \tag{25}$$

which in turn yields

$$H_{\mathrm{MF}}(\beta) = H_Q - \frac{\kappa_0(\beta)}{2}f^2 + \frac{1}{\beta}\log Z_X(\beta)\,\mathbb{I}. \tag{26}$$

Notably this correction to $H_{\mathrm{MF}}(\beta)$ is entirely *classical* [1]. This is hardly surprising, but emphasises that quantum effects *must* stem from non-commutativity. A truly quantum theory of thermodynamics therefore only makes meaningfully distinct predictions when $[H_Q, f] \neq 0$. In this case however, the noise enters through a noncommuting operator inside $\mathcal{T}_\tau$, rendering the question of averaging highly non-trivial. In the next section, we attack this problem directly, deriving conditions under which $H_{\mathrm{MF}}(\beta)$ possesses a closed form.

GENERATING-FUNCTION STRUCTURE OF THE INFLUENCE OPERATOR

Quenched propagator and Gaussian resummation

The starting point is the quenched propagator introduced in Sec. . In the imaginary-time interaction picture with respect to H_Q, the unnormalised reduced equilibrium operator takes the form

$$\begin{aligned}\bar{\rho}_Q(\beta) &= e^{-\beta H_Q}\,\langle W_\xi(\beta)\rangle_\xi,\\ W_\xi(\beta) &\equiv \mathcal{T}_\tau \exp\left[-\int_0^\beta d\tau\,\xi(\tau)\,\tilde{f}(\tau)\right],\end{aligned} \tag{27}$$

where $\tilde{f}(\tau) = e^{\tau H_Q} f e^{-\tau H_Q}$ and the noise covariance is $\langle\xi(\tau)\xi(\tau')\rangle_\xi = K(\tau - \tau')$.

Because the bath is Gaussian, only the second cumulant of ξ is non-vanishing. Time-ordering introduces a non-trivial operator structure, but the key identity—proved in Appendix A—is that the Wick resummation of all pairings exponentiates exactly:

$$\langle W_\xi(\beta)\rangle_\xi = \exp\left[\frac{1}{2}\int_0^\beta d\tau\int_0^\beta d\tau'\, K(\tau - \tau')\,\mathcal{T}_\tau\big[\tilde{f}(\tau)\tilde{f}(\tau')\big]\right]. \tag{28}$$

The proof is a direct application of Wick's theorem to the Dyson expansion of W_ξ: at order $2p$ in the expansion, there are $(2p)!/(2^p p!)$ distinct Wick pairings, each weighted by the two-point kernel $K(\tau_i - \tau_j)$. The prefactors from the Dyson expansion precisely cancel the pairing multiplicity, leaving only the singly-connected pairs.

We can exploit the symmetry $K(\tau - \tau') = K(\tau' - \tau)$ to fold the square domain onto the triangular one (see Appendix A), such that the exponent simplifies to

$$\Delta(\beta) \equiv \int_0^\beta d\tau\int_0^\tau d\tau'\, K(\tau - \tau')\,\tilde{f}(\tau)\tilde{f}(\tau'), \tag{29}$$

and the averaged reduced state takes the product form

$$\bar{\rho}_Q(\beta) = e^{-\beta H_Q}\, e^{\Delta(\beta)}. \tag{30}$$

The influence exponent $\Delta(\beta)$ is quadratic in $\tilde{f}$ and depends on the bath exclusively through the two-point kernel K. Here the absence of higher-order bath objects is not an approximation, but an exact consequence of the Gaussian nature of the environment.

The ordered Green function and frequency compression

At this stage, the operator Δ is still formally defined as an integral over the bath degrees of freedom. There is however a particularly satisfying interpretation of this operator in which the effect of the bath is readily understood. To make this explicit, we work in the eigenbasis of H_Q. Let $H_Q|i\rangle = E_i|i\rangle$, so that the matrix elements of the interaction-picture operator are

$$\tilde{f}_{ij}(\tau) = \langle i|\tilde{f}(\tau)|j\rangle = f_{ij}\, e^{\omega_{ij}\tau}, \qquad \omega_{ij} \equiv E_i - E_j. \quad (31)$$

Substituting into Eq. (29) and expanding in the eigenbasis, the influence exponent takes the spectral form

$$\Delta(\beta) = \sum_{i,\ell} \Delta_{i\ell}\, |i\rangle\langle\ell|, \quad (32)$$

where evaluating the matrix elements of the operator product yields

$$\Delta_{i\ell} = \sum_j f_{ij}\, f_{j\ell}\, G^{>}(\omega_{ij}, \omega_{j\ell}), \quad (33)$$

where all bath and temperature dependence has been absorbed into the *ordered Green function*

$$G^{>}(\omega_1, \omega_2) \equiv \int_0^\beta d\tau \int_0^\tau d\tau'\, K(\tau - \tau')\, e^{\omega_1\tau + \omega_2\tau'}. \quad (34)$$

In standard finite-temperature perturbation theory, analogous time-ordered integrals are evaluated using discrete Matsubara frequencies. Here, the integration instead projects the bath response directly onto the Laplace frequencies set by the system's own energy differences; the superscript $>$ records that $\tau > \tau'$ throughout the integration domain.

The double integral in Eq. (34) collapses to a one-dimensional Laplace transform upon changing variables to $u = \tau - \tau'$ and $v = \tau'$:

$$G^{>}(\omega_1, \omega_2) = \frac{e^{\beta\omega_1}\mathcal{K}(\omega_2) - \mathcal{K}(\omega_1)}{\omega_1 + \omega_2}, \quad (35)$$

where

$$\mathcal{K}(\omega) \equiv \int_0^\beta du\, K(u)\, e^{\omega u} \quad (36)$$

is the bare Laplace transform of the bath kernel. Eq. (35) is an exact closed form valid for all $\omega_1 + \omega_2 \neq 0$. Importantly, it implies that for any transition between levels in the reduced system, the entirety of the bath's influence is encoded by $\mathcal{K}$ evaluated at the transition frequency.

The resonant channel $\omega_1 + \omega_2 = 0$ requires separate treatment. Taking the limit carefully yields

$$R(\omega) \equiv G^{>}(\omega, -\omega) = \int_0^\beta (\beta - u)\, K(u)\, e^{\omega u}\, du, \quad (37)$$

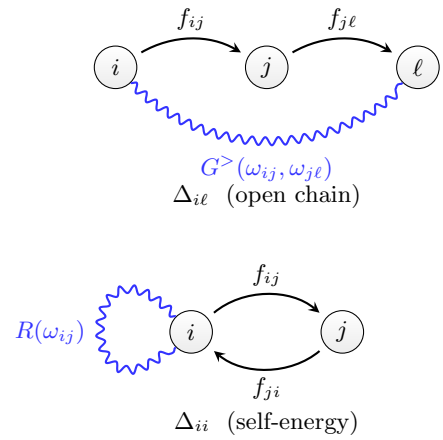


FIG. 1. Diagrammatic structure of the influence exponent $\Delta(\beta)$ in the H_Q eigenbasis. *Top*: The off-diagonal element $\Delta_{i\ell}$ (open chain) sums intermediate sites j, weighted by the bath propagator $G^{>}(\omega_{ij}, \omega_{j\ell})$ (wavy line) and two coupling insertions f (arrows). Energy is not conserved at the intermediate vertex. *Bottom*: The diagonal element Δ_{ii} (self-energy) is a closed loop $i \to j \to i$; the resonant condition $\omega_1 + \omega_2 = 0$ is satisfied, and the bath propagator collapses to $R(\omega_{ij})$.

where the factor $(\beta - u)$ is the available imaginary time after a bath correlation at separation u; the resonant kernel thus accumulates bath memory over the full thermal interval, regulated by the finiteness of β.

The structure of Eq. (33) admits a pleasingly simple diagrammatic reading, illustrated in Fig. 1. To isolate the physically distinct effects of the bath, we partition the influence exponent into its secular (diagonal) and mixing (off-diagonal) components, $\Delta = \Delta_\parallel + \Delta_\perp$.

The secular sector $\Delta_\parallel \equiv \sum_i \Delta_{ii}|i\rangle\langle i|$ singles out the *closed loops* $i \to j \to i$, for which $\omega_{ij} + \omega_{ji} = 0$ identically. The resonance condition forces the bath propagator onto $R(\omega_{ij})$, the object that accumulates bath memory over the whole thermal interval. The diagonal elements are thus the exact, non-perturbative *self-energy* dressing of the i-th energy level by all levels j it is coupled to:

$$\Delta_{ii} = \sum_j |f_{ij}|^2\, R(\omega_{ij}). \quad (38)$$

Conversely, the mixing sector $\Delta_\perp \equiv \sum_{i\neq\ell} \Delta_{i\ell}|i\rangle\langle\ell|$ describes *open chains* $i \to j \to \ell$, where energy is not conserved at the intermediate vertex. Using the closed form in Eq. (35), these components can be expressed directly in terms of the Laplace-transformed bath kernel $\mathcal{K}(\omega)$:

$$\Delta_{i\ell} = \frac{1}{\omega_{i\ell}} \sum_j f_{ij} f_{j\ell} \left[e^{\beta\omega_{ij}}\mathcal{K}(\omega_{j\ell}) - \mathcal{K}(\omega_{ij}) \right] \quad (i \neq \ell), \quad (39)$$

which encodes the coherent transitions between distinct energy levels mediated by the asymmetric fluctuations of the bath.

KMS symmetry and the symmetrised reduced state

The bath kernel $K(\tau)$ satisfies the KMS periodicity condition $K(\tau) = K(\beta-\tau)$, which in Laplace space reads

$$\mathcal{K}(-\omega) = e^{-\beta\omega}\,\mathcal{K}(\omega). \tag{40}$$

Substituting into the closed form Eq. (35) and simplifying gives a corresponding symmetry of the ordered Green function:

$$G^{>}(-\omega_2,-\omega_1) = e^{-\beta(\omega_1+\omega_2)}\,G^{>}(\omega_1,\omega_2). \tag{41}$$

Because transition frequencies telescope, $\omega_{ij}+\omega_{j\ell}=\omega_{i\ell}$, applying Eq. (41) to the matrix-element formula Eq. (33) immediately yields the *detailed-balance* constraint

$$\Delta_{\ell i} = e^{-\beta\omega_{i\ell}}\,\Delta^*_{i\ell}, \tag{42}$$

or in operator form $\Delta^\dagger = \Pi\,\Delta\,\Pi^{-1}$ where $\Pi \equiv e^{-\beta H_Q}$. Although Δ itself is not self-adjoint, this relation is a precise constraint: the product $\Pi\Delta$ is Hermitian.

This symmetry allows us to recast the unnormalised reduced state Eq. (30) into a manifestly positive, Hermitian form. Writing the bare Gibbs factor as $\Pi \equiv e^{-\beta H_Q}$ gives $\bar\rho_Q = \Pi\,e^{\Delta}$. By splitting $\Pi = \Pi^{1/2}\Pi^{1/2}$ and inserting the identity $\Pi^{-1/2}\Pi^{1/2} = \mathbf{1}$, we symmetrically distribute the bare weight:

$$\bar\rho_Q = \Pi^{1/2}\big(\Pi^{1/2}\,e^{\Delta}\,\Pi^{-1/2}\big)\Pi^{1/2}. \tag{43}$$

Applying the operator identity $A\,e^{B}A^{-1} = \exp(ABA^{-1})$ to the inner bracket yields

$$\bar\rho_Q = \Pi^{1/2}\,e^{S}\,\Pi^{1/2}, \qquad S \;\equiv\; \Pi^{1/2}\,\Delta\,\Pi^{-1/2}, \tag{44}$$

where S is the *symmetrised influence operator*. Its matrix elements evaluate to $S_{i\ell} = e^{-\beta\omega_{i\ell}/2}\Delta_{i\ell}$, which by Eq. (42) satisfy $S^*_{i\ell} = S_{\ell i}$. Thus, the modified exponent S is strictly self-adjoint. The partition function of the reduced system follows by cyclicity of the trace:

$$Z_Q \;\equiv\; \mathrm{Tr}[\bar\rho_Q] = \mathrm{Tr}[\Pi\,e^{S}] = \sum_i e^{-\beta E_i}\,(e^{S})_{ii}. \tag{45}$$

Since S is self-adjoint, $(e^S)_{ii} \geq e^{S_{ii}} = e^{\Delta_{ii}}$ by Jensen's inequality applied to the spectral weights $|\langle i|n\rangle|^2$, giving the bound

$$Z_Q \;\geq\; \sum_i e^{-\beta E_i+\Delta_{ii}}, \tag{46}$$

where the right-hand side is a Boltzmann sum with bath-shifted energies and equality holds when S is diagonal (i.e. when H_Q and Δ share an eigenbasis). The full result $Z_Q = \mathrm{Tr}[\Pi e^S]$ captures all level-mixing corrections through the off-diagonal elements of S.

THE HAMILTONIAN OF MEAN FORCE

Closure as an operator-algebra statement

With this, we may now give the general expression for the Hamiltonian of Mean-Force.

$$-\beta H_{\mathrm{MF}}(\beta) = \log\big(e^{-\beta H_Q}\,e^{\Delta(\beta)}\big). \tag{47}$$

If H_Q and Δ commuted, the logarithm would trivially give $-\beta H_Q + \Delta$. In general they do not, and the Baker–Campbell–Hausdorff (BCH) expansion

$$\log(e^{A}e^{B}) = A+\Phi(\mathrm{ad}_A)\,B+O(B^2), \qquad \Phi(x) \equiv \frac{x}{1-e^{-x}}, \tag{48}$$

generates a tower of commutators of $A = -\beta H_Q$ acting on $B = \Delta$. To understand the structure of these corrections, we invoke the partition of the influence exponent into its secular and mixing parts, $\Delta = \Delta_\parallel + \Delta_\perp$. The structural partitioning of these mean-force corrections is illustrated in Fig. 2. Because the secular piece $\Delta_\parallel$ strictly commutes with A, its adjoint action is annihilated ($\mathrm{ad}_A(\Delta_\parallel) = 0$).

Consequently, the generating function evaluated at $x = 0$ yields $\Phi(0) = 1$, allowing the parallel component to push straight through the linear BCH expansion:

$$-\beta H_{\mathrm{MF}} \approx \big(-\beta H_Q + \Delta_\parallel\big) + \Phi(\mathrm{ad}_{-\beta H_Q})\,\Delta_\perp + O(\Delta^2). \tag{49}$$

The secular zeroth-order shift $\Delta_\parallel$ thus groups directly with the bare system (reminiscent of the weak-coupling Lamb shift). Crucially however, while it acts geometrically like a classical energy shift, this term is still fundamentally *quantum* in origin because it is weighted by the bath Green functions, dynamically entangling the bare spectrum with the full spectral density of the environment. We can understand this directly by reference to Eq. (38). By defining a bath-dressed coupling operator $\tilde f$ with matrix elements $\tilde f_{ij} = f_{ij}\sqrt{R(\omega_{ij})}$, the secular exponent takes the exact quadratic form $\Delta_\parallel = (\tilde f\tilde f^\dagger)_{\mathrm{diag}}$. In the macroscopic limit where the system commutes with its coupling ($[H_Q, f] = 0$), the matrix f is entirely diagonal in the energy basis ($\omega_{ij} = 0$), and this dressed square collapses precisely to $\Delta_\parallel = \beta\lambda f^2$, where $R(0) = \beta\lambda$ is the explicit reorganisation energy defined in Eq. (23). The zeroth order BCH term therefore exactly recovers the classical commuting shift $H_{\mathrm{MF}} = H_Q - \lambda f^2$ of Eq. (26), and at zeroth order the non-commuting correction can be obtained by replacing $f \to \tilde f$. The remaining coherence-generating effects are then driven exclusively by the action of Φ on the perpendicular, off-diagonal open-chain fragments $\Delta_\perp$.

There is a saitsfyingly clean expression for these non-commuting corrections in the system energy eigenbasis. The off-diagonal basis operators $|i\rangle\langle j|$ are exact eigenoperators of the adjoint map: $\mathrm{ad}_{-\beta H_Q}|i\rangle\langle j| =$

$-\beta[H_Q, |i\rangle\langle j|] = -\beta\omega_{ij}|i\rangle\langle j|$. Consequently, the entire infinite tower of nested commutators generated by $\Phi(\mathrm{ad}_{-\beta H_Q})$ trivially evaluates to a simple scalar multiplication for each matrix element of the perpendicular fragment:

$$\Phi(\mathrm{ad}_{-\beta H_Q})\Delta_\perp = \sum_{i\neq j} \Phi(-\beta\omega_{ij})\,\Delta_{ij}|i\rangle\langle j|, \tag{50}$$

where $\Phi(x) = x/(1-e^{-x})$. The off-diagonal elements of the linear-in-Δ effective Hamiltonian are therefore simply the bare bath-induced transitions Δ_{ij}, parametrically reweighted by the thermal asymmetry factor $\Phi(-\beta\omega_{ij}) = -\beta\omega_{ij}/(1-e^{\beta\omega_{ij}})$.

This scalar evaluation exhibits a manifestly physical symmetry. If we reconstruct the effective matrix element using the symmetrised influence operator $S_{ij} = e^{-\beta\omega_{ij}/2}\Delta_{ij}$, the thermal asymmetry factor exactly absorbs the detailed-balance twist. The (off-diagonal) components on the effective Hamiltonian are therefore simply the bare bath-induced transitions Δ_{ij}, parametrically reweighted by the thermal asymmetry factor $\Phi(-\beta\omega_{ij}) = -\beta\omega_{ij}/(1-e^{\beta\omega_{ij}})$:

$$\begin{aligned}(H_{\mathrm{MF}})_{ij} &= -\frac{1}{\beta}\Phi(-\beta\omega_{ij})e^{\beta\omega_{ij}/2}S_{ij} \\ &= -\frac{\omega_{ij}/2}{\sinh(\beta\omega_{ij}/2)}\,e^{-\beta\omega_{ij}/2}\Delta_{ij}.\end{aligned} \tag{51}$$

Because $x/\sinh(x)$ is an explicitly even geometric function of the energy gap ω_{ij}, Eq. (51) guarantees that the linearly truncated effective Hamiltonian is perfectly Hermitian. Moreover, this exact thermal weighting reveals that bath-induced coherent mixing between levels is heavily exponentially suppressed for high-frequency transitions ($|\beta\omega_{ij}| \gg 1$), dynamically localising the leading mean force corrections to the thermodynamic energy shell.

This fortuitous cancellation reveals a deep structural connection between quantum thermodynamics and Lie algebra. The fundamental detailed balance condition of the bath imposes a thermal asymmetry $e^{-\beta\omega}$ on the bare transition amplitudes. Conversely, the BCH expansion—a purely abstract property of non-commuting logarithms—generates the parametric weighting $\Phi(-\beta\omega) \propto (1-e^{\beta\omega})^{-1}$. The fact that the Lie-algebraic geometry of the thermal exponential perfectly mirrors - and thereby absorbs - the thermodynamic detailed-balance twist of the environment is striking, and illustrative of the high degree of mathematical harmonisation underlying the mean-force formulation.

The foregoing parametric logic naturally generalises directly to all higher orders of the BCH expansion. Because the basis operators $|i\rangle\langle j|$ are strict eigenoperators of the adjoint map, the evaluation of any highly nested Lie commutator string spanning $O(\Delta^n)$ simply produces ordinary matrix products of the abstract fragments $\Delta_{ik_1}\Delta_{k_1k_2}\ldots\Delta_{k_{n-1}j}$, completely bypassing the need for functional derivatives. Over the energy eigenbasis, the entire non-linear algebraic sum reduces to evaluating n-step discrete pathways through the bare spectrum, where each composite transition amplitude is identically parametrically weighted by a universal, symmetric multivariate generalisation of Φ determined solely by Lie combinatorics.

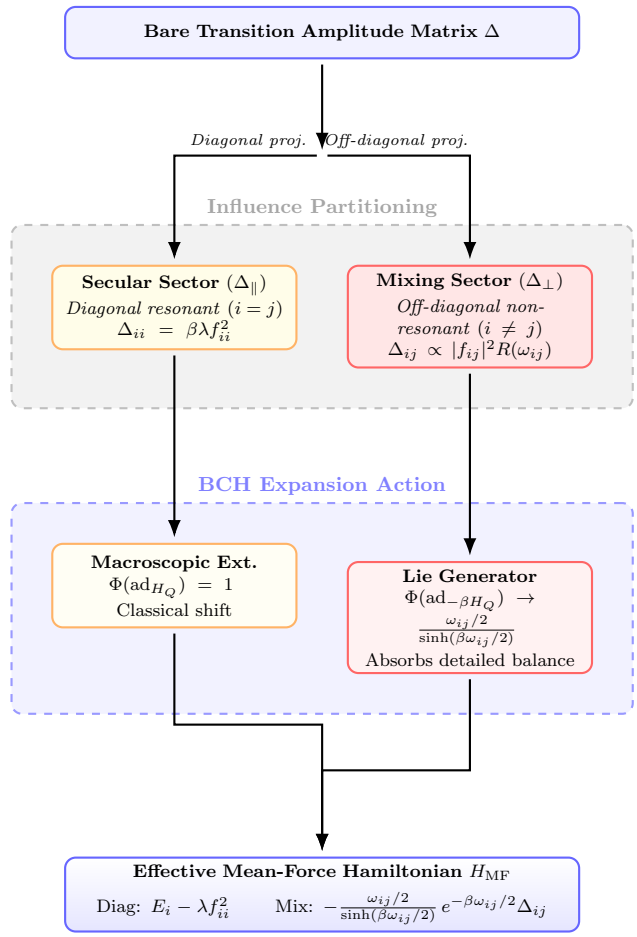


FIG. 2. Structural decomposition of the linear-order Mean-Force Hamiltonian. The parallel (secular) sector generates classical Lamb-like commuting phase shifts, entirely isolated from detailed balance effects. Concurrently, the perpendicular (mixing) sector passes through the Baker–Campbell–Hausdorff non-commutative generator, producing a geometric Lie scaling that perfectly absorbs the detailed balance thermal twist.

While this geometric pathing approach is exceptionally suggestive—and likely a numerically robust framework for resolving perturbative components—the abstract operator structure is more directly revealing of the core algebraic symmetries. Because the adjoint action obeys the rule $\mathrm{ad}_{H_Q}(f_n f_m) = f_{n+1}f_m + f_n f_{m+1}$ (where $f_n \equiv \mathrm{ad}_{H_Q}^n f$), every nested BCH commutator stays within the algebra generated by the adjoint chain of the coupling.

This ensures that even high-order nonlinear corrections introduce *no new bath objects*: the environment continues to enter exclusively through the fundamental spectral functions $\mathcal{K}(\omega)$ and $R(\omega)$, with coefficients determined solely by universal Lie combinatorics.

The closure content of $H_{\mathrm{MF}}(\beta)$ is therefore the following: the entire operator lives in the finite-dimensional algebra generated by $\{H_Q, \Delta\}$, with Δ quadratic in f and the bath entering exclusively through the two spectral objects $\mathcal{K}(\omega)$ and $R(\omega)$. Whether this algebra is itself finite-dimensional - and hence whether H_{MF} admits a closed form in a restricted operator family - depends on the structure of f and H_Q This statement leads to an important corollary. For continuous variable systems $H_Q = p^2/2m + V(x)$, if the potential $V(x)$ is a polynomial of degree greater than 2, the adjoint chain $\{f_n\}$ generally does not close in any finite-dimensional Lie algebra; instead, it generates the full universal enveloping algebra. Consequently, no finite representation of the mean force exists within the original operator class. This obstruction is analogous (in spirit) to the Groenewold–Van Hove no-go theorem on consistent quantisation of polynomial observables: the failure to close the Lie algebra under adjoint action corresponds to the impossibility of mapping the quantum evolution of non-quadratic potentials to a finite phase space flow. For such potentials, there is no *local operator* that exactly describes the reduced equilibrium state.

From a practical standpoint, the closure of the adjoint-chain is not necessary for an arbitrarily accurate approximation of $H_{\mathrm{MF}}(\beta)$, as the BCH logarithm $\log(e^{-\beta H_Q} e^{\Delta})$ still yields a controlled expansion. From this algebraic perspective, the closure problem offers a unifying view of strong-coupling approximations. Numerical schemes such as the polaron transformation [30, 56], hierarchical equations of motion [37], and chain-mapping/DMRG techniques [39] can be understood as distinct choices of a truncated operator family $\mathcal{A}$. The polaron frame corresponds to dressing $\mathcal{A}$ with coherent displacements, HEOM truncates the memory depth of the influence functional (dual to the operator degree in the adjoint chain), and chain mappings truncate the spatial extent of the harmonic bath. The 'closure' problem is thus equivalent to identifying a low-dimensional subalgebra that approximately contains the logarithm of the quenched propagator.

SPIN-BOSON MODEL

We now put the machinery of Secs. – to work in the prototypical case of an open quantum system - the Spin-Boson model. This couples a single spin to the environment, and its $\mathfrak{su}(2)$ algebra is rich enough to generate non-trivial corrections, while being simple enough to yield easily tractable results. Critically however, this latter feature is *convenient* for our purposes, and not reflective of a limitation of the general framework. As illustrated in Fig. 3, the exact qubit mean-force state $\mathbf{r}$ lives in a three-dimensional Bloch sphere, but is geometrically locked to the *coupling plane* (shaded) spanned by the system axis $\hat{\mathbf{n}}_s$ and the interaction axis $\hat{\mathbf{f}}$. The orientation of this plane is defined by the interaction phase ϕ, which directly determines the phase of the induced coherences in the reduced state.

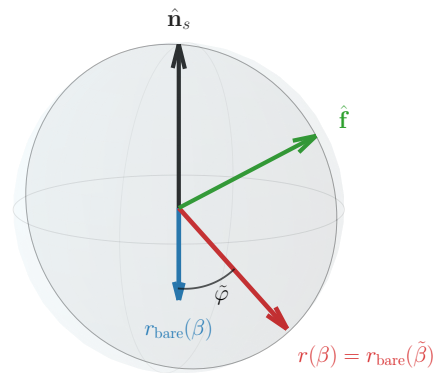


FIG. 3. 3D geometric overview of the qubit mean-force state. The bare system is defined by the axis $\hat{\mathbf{n}}_s$ at the north pole, with the bare thermal state vector pointing towards the south pole with length $r(\beta)$. The interaction axis $\hat{\mathbf{f}}$ is also shown. The exact reduced state $\mathbf{r}$ has magnitude $r(\tilde{\beta})$ and is tilted by the effective angle $\tilde{\varphi}$ relative to the bare state. The state remains geometrically locked to the shaded coupling plane.

We first define our system Hamiltonian and coupling via

$$H_Q = \frac{\omega_q}{2} \hat{\mathbf{n}}_s \cdot \boldsymbol{\sigma}, \qquad f = \hat{\mathbf{f}} \cdot \boldsymbol{\sigma}, \tag{52}$$

where $\hat{\mathbf{n}}_s$ and $\hat{\mathbf{f}}$ are unit vectors defining the system and coupling axes, respectively. The overall coupling strength g is omitted in the following derivation for clarity and can be re-introduced as a global scaling factor where needed. For the derivation it is convenient to work in the eigenbasis of H_Q (so $\hat{\mathbf{n}}_s = \hat{\mathbf{z}}$). The final formulas are then rewritten covariantly in terms of $\hat{\mathbf{n}}_s$ and $\hat{\mathbf{f}}_\perp$.

The influence exponent may easily be calculated directly, using $\tilde{f}(\tau) = e^{\tau H_Q} f e^{-\tau H_Q}$. In the qubit energy basis ($\hat{\mathbf{n}}_s = \hat{\mathbf{z}}$), we decompose the unit coupling vector as $\hat{\mathbf{f}} = (s\cos\phi, s\sin\phi, c)$, where $c \equiv \cos\theta$ and $s \equiv \sin\theta$ denote the longitudinal and transverse projections relative

to the system axis, and ϕ is the azimuthal angle. Writing

$$\begin{aligned} f &= c\sigma_z + f_-\sigma_+ + f_+\sigma_-, \\ f_\pm &\equiv s e^{\pm i\phi}, \end{aligned} \tag{53}$$

gives

$$\tilde{f}(\tau) = c\sigma_z + f_- e^{\omega_q \tau}\sigma_+ + f_+ e^{-\omega_q \tau}\sigma_-, \tag{54}$$

meaning $\Delta(\beta)$ is determined solely by the bath response kernels at the frequencies $0, \pm\omega_q$. Consequently, all bath dependence in Δ is compressed to scalar kernel data evaluated at the same frequencies. Using the notation of Sec. , the required bath objects are $\mathcal{K}(0)$, $\mathcal{K}(\pm\omega_q)$, and $R(0)$, $R(\pm\omega_q)$.

For a single qubit, the operator algebra closes almost trivially. Using $\sigma_+\sigma_- = (\mathbb{I} + \sigma_z)/2$, $\sigma_-\sigma_+ = (\mathbb{I} - \sigma_z)/2$, and $\sigma_\pm^2 = 0$, one finds

$$\begin{aligned} \tilde{f}(\tau)\tilde{f}(\tau') = &\left[c^2 + s^2\cosh\left(\omega_q(\tau-\tau')\right)\right]\mathbb{I} \\ &+ s^2\sinh\left(\omega_q(\tau-\tau')\right)\sigma_z \\ &+ c f_-\left(e^{\omega_q\tau'} - e^{\omega_q\tau}\right)\sigma_+ \\ &+ c f_+\left(e^{-\omega_q\tau} - e^{-\omega_q\tau'}\right)\sigma_-. \end{aligned} \tag{55}$$

Using the response kernels, the entirety of the bath's influence is compressed into two real, temperature-dependent response kernels: a longitudinal (resonant) kernel Σ_z and a transverse (non-resonant) kernel $\Sigma_\perp$, defined as

$$\begin{aligned} \Sigma_z &\equiv \frac{1}{2}\left[R(\omega_q) - R(-\omega_q)\right], \\ \Sigma_\perp &\equiv \frac{4}{\omega_q}\left[\cosh\left(\frac{\beta\omega_q}{2}\right)\mathcal{K}(0) - e^{-\beta\omega_q/2}\mathcal{K}(\omega_q)\right]. \end{aligned} \tag{56}$$

To expose the coupling dependence cleanly, we reintroduce an explicit dimensionless strength g through $K(u) = g^2 K_0(u)$, so that all response kernels scale as g^2. In the interests of brevity, we shall subsume this factor into the definition of parameters, except in those cases where physically relevant effects depend on it.

The reduced system's $\mathfrak{su}(2)$ algebra means both the influence exponent and the final reduced state may be calculated exactly. After some algebraic gymnastics (done without harness in App C), the exact reduced density matrix ρ_Q takes the form:

$$\rho_Q = \frac{1}{2}\begin{pmatrix} 1 + m_z & m_\perp e^{-i\phi} \\ m_\perp e^{i\phi} & 1 - m_z \end{pmatrix}. \tag{57}$$

Both the definition and the interpretation of the Bloch vector component magnetisations $(m_z, m_\perp)$ relies on several intermediate parametrisations. First and most importantly, we have the *influence magnitude*

$$\chi = s\sqrt{c^2\Sigma_\perp^2 + s^2\Sigma_z^2}, \tag{58}$$

which - as we will see shortly - acts as the essential determinant for the different regimes of behaviour for the system. It is equivalent to the *norm* of the influence exponent $\|\Delta(\beta)\|^2 \equiv \chi^2$, up to the sign of $\sin(\theta)$. This sign has significant consequences, but like g we shall work under the assumption $\sin(\theta) > 0$ except for those cases where this sign is relevant. The reduced state's Bloch vector radius r is then given by

$$r = |\tanh\chi| = \sqrt{m_z^2 + m_\perp^2}. \tag{59}$$

If we define the parameters

$$\Theta \equiv \operatorname{arctanh}(\gamma s^2\Sigma_z) - \frac{\beta\omega_q}{2}, \tag{60}$$

$$\tan\eta \equiv -\cot\theta\frac{\Sigma_\perp}{\Sigma_z}, \tag{61}$$

$$\gamma = \tanh\chi/\chi, \tag{62}$$

then the Bloch vector components are expressed via

$$m_z = \tanh\Theta, \tag{63}$$

$$m_\perp = \tan\eta\left(m_z\cosh\frac{\beta\omega_q}{2} + \sinh\frac{\beta\omega_q}{2}\right), \tag{64}$$

$$\tan\varphi = m_\perp/m_z. \tag{65}$$

From this, we see that the reduced state is confined to the common plane of $\hat{\mathbf{n}}_s$ and $\hat{\mathbf{f}}$, with its radius given by r and a *tilt-angle* φ away from the bare system's axis $\hat{\mathbf{n}}_s$. The parameter $\tan\eta$ then encodes the anisotropy of the bath's influence as a function of θ. In particular, when $\theta < \pi/2$, $\tan\eta < 0$ and the state tilts toward the ground-state hemisphere. for $\theta > \pi/2$, the sign of $\tan\eta$ reverses and the state is dragged toward the *excited*-state hemisphere, corresponding to a bath-induced population inversion relative to the bare Hamiltonian eigenbasis. The special cases $\cos(\theta) = 0$ (purely transverse coupling) has $\tan\eta = 0$ identically.

The importance of this anisotropy is made clear when we consider the interpretation of Θ, which serves as an effective *renormalisation* of temperature. We may justify this interpretation by first noting the special case of a purely transverse coupling $\theta = \frac{\pi}{2}$. In this case the influence operator aligns with $\hat{n}_s$ the reduced system is simply a copy of the bare system with $\frac{\beta\omega_q}{2} \to \Theta$. This immediately implies that we may describe the system via an *effective* temperature $\tilde{\beta}$:

$$\beta \to \tilde{\beta} = \frac{2\Theta}{\omega_q}. \tag{66}$$

This temperature renormalisation is a universal feature holding at any coupling angle θ. Critically though, whether this renormalisation maps to a higher or lower temperature depends on the *relative sign* of Θ and $\sin(\theta)$. Switching the sign of either corresponds to $m_z \to -m_z$ and population inversion. The only contribution of the

coupling angle beyond this is simply to *rotate* the temperature renormalised thermal state relative to the bare orientation. We can see this most clearly by returning to Eq.(64). Defining the factor

$$\kappa = \sinh(\operatorname{arctanh}(\gamma s^2 \Sigma_z)), \tag{67}$$

after rearrangement, $m_\perp$ takes the form

$$m_\perp = -\kappa \frac{\Sigma_\perp}{\Sigma_z} \tan(\theta - \frac{\pi}{2}) \cosh^{-1}\left(-\frac{\tilde{\beta}\omega_q}{2}\right). \tag{68}$$

This formulation provides a natural definition for the effective longitudinal and transverse magnetisations of the dressed qubit:

$$\tilde{m}_z \equiv \frac{\Sigma_z \sinh\left(-\frac{\tilde{\beta}\omega_q}{2}\right)}{\tilde{r}}, \qquad \tilde{m}_\perp \equiv \frac{\kappa \Sigma_\perp}{\tilde{r}}. \tag{69}$$

where $\tilde{r}(\beta) = r(\tilde{\beta})$ is the renormalised radius. This means the renormalised angle at $\tilde{\beta}$ is given by

$$\tan\tilde{\varphi} = \frac{\kappa \Sigma_\perp \tan(\theta - \frac{\pi}{2})}{\Sigma_z \sinh \frac{\tilde{\beta}\omega_q}{2}}. \tag{70}$$

In this representation, the effective tilt angle $\tilde{\varphi}$ reflects the competition between the resonant reinforcement of the ground state and the non-resonant generation of coherence. A non-zero $\tilde{\varphi}$ signifies that the bath's influence has rotated the reduced state away from the bare energy basis, endowing the equilibrium state with apparent "vacuum-induced" coherences with magnitude $|\tilde{m}_\perp|$. For $\theta > \pi/2$ the tilt is toward the excited-state direction, so the bath-induced rotation and any strong-coupling purity enhancement act together to favour excited-state population: the reduced state can be effectively population-inverted.

PHYSICAL REGIMES OF THE MEAN-FORCE STATE

Even in this simple case, the physics encoded in the mean force state by its dependence on temperature and bath coupling is startlingly rich. To expose this, we first commit to a specific bath spectral density for numerics. We choose the Ohmic spectral density

$$J(\omega) = \alpha\, \omega\, e^{-\omega/\omega_c}, \qquad \omega \in [0, 2\omega_q], \tag{71}$$

with $\alpha = 1$ and $\omega_c = 5\omega_q$. Unless otherwise stated, we further specialise to $\theta = \pi/4$ throughout. We emphasise here that the choice of spectral shape is not fundamental, as it only controls the way in which χ runs with temperature.

This apparently innocuous expression encodes the entire non-perturbative recombination of bare Hamiltonian and influence, and functions as an *order* parameter for the crossover between weak and strong coupling regimes. The reduced equilibrium state transitions between these regimes as χ passes through unity,

$$\chi(\beta, g, \theta) \sim 1 \qquad \Longleftrightarrow \qquad g^2 \sim \chi_0(\beta, \theta)^{-1}. \tag{72}$$

This crossover condition $\chi(\beta, g, \theta) = 1$ therefore defines a coupling-dependent temperature scale

$$g_\star(\beta) = \chi_0(\beta, \theta)^{-1/2}, \tag{73}$$

and an equivalent crossover temperature $\beta_\star(g)$. Notably, these dual crossover values are an exact property of the solution.

The function $\gamma(\chi)$ is elementary but worth dwelling on. It is monotone decreasing from $\gamma(0) = 1$ to $\gamma(\infty) = 0$ with its inflection near $\chi \approx 1.2$, and admits clean asymptotic forms:

$$\chi \ll 1: \quad \gamma \simeq 1 - \tfrac{\chi^2}{3} \quad \text{(weak coupling)}, \tag{74}$$

$$\chi \gg 1: \quad \gamma \simeq \chi^{-1} = (g^2 \chi_0)^{-1} \quad \text{(ultrastrong)}. \tag{75}$$

In the weak-coupling regime the state tracks the bare Gibbs thermal state with $O(g^2)$ corrections, as expected. Notably, as shown in Fig. 4, the crossover between regimes depends on *both* temperature and coupling strength. This means that two distinct regimes emerge in the limit of scaling g and β, wholly dependent on which side of the crossover is taken.

Notably, these critical temperatures and couplings have an *involutive* relationship, in the sense that

$$\beta_\star(g_\star(\beta)) = \beta, \quad g_\star(\beta_\star(g)) = g. \tag{76}$$

This relationship between temperature and coupling means the crossover can be interpreted as the *self-dual* line in the (β, g) plane, as explored in Fig. 5. This line represents a coupling-dependent temperature scale $\beta_\star(g)$ (or equivalently $g_\star(\beta)$), where different parameter scaling relationships apply.

State purity

The purity of the qubit mean-force state is $\mathcal{P} = \frac{1}{2}(1 + r^2)$ with $r^2 = |\mathbf{r}|^2$. From this we can immediately establish a universal result:

> **Bath always purifies.** *For any positive spectral density $J(\omega) \geq 0$ and any $\beta > 0$: (i) $r^2(g, \beta)$ is strictly increasing in g for $g \geq 0$, and (ii) for all $g > 0$,*
>
> $$r^2(g, \beta) \;>\; r^2_{\text{bare}} \;\equiv\; \tanh^2\left(\frac{\beta\omega_q}{2}\right), \tag{77}$$
>
> *with equality only at $g = 0$.*

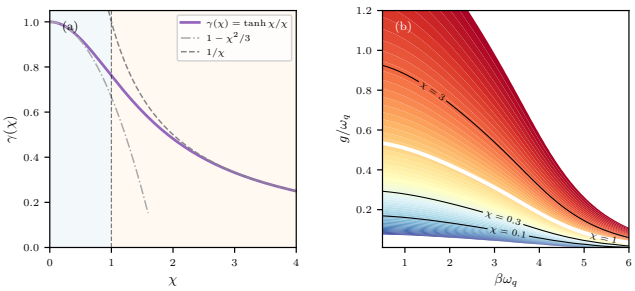


FIG. 4. Universal crossover for the qubit mean-force state. **(a)** Crossover function $\gamma(\chi) = \tanh\chi/\chi$ interpolating between the weak-coupling approximant $1-\chi^2/3$ and the ultrastrong form $1/\chi$. Shaded regions mark the weak (blue) and strong (orange) regimes separated by $\chi = 1$. **(b)** Dimensionless coupling $\chi(\beta, g) = g^2\chi_0(\beta)$ as a contour map. Vertical trajectories (fixed g) will crossover at its corresponding $\beta_\star(g)$.

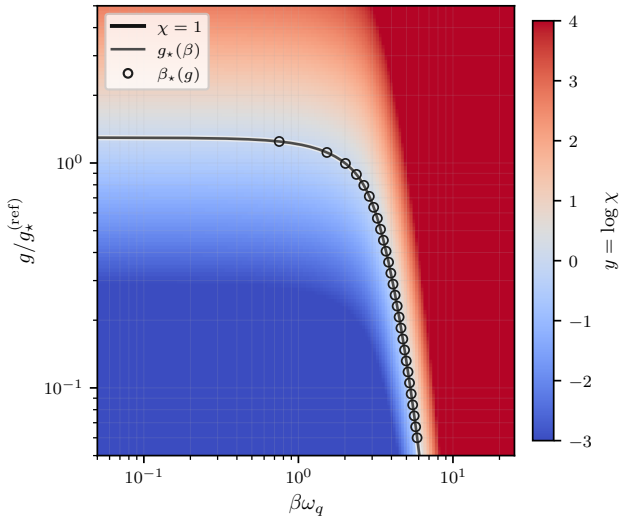


FIG. 5. Involutive crossover geometry generated by $\chi(\beta, g) = g^2\chi_0(\beta)$. The heatmap shows $y = \log\chi$ in the (β, g) plane. The solid black line marks the crossover locus $\chi = 1$, which can be constructed either as a coupling-dependent temperature $\beta_\star(g)$ (circles) or a temperature-dependent coupling $g_\star(\beta)$ (solid grey line). This involutive relationship defines a partition between the weak-coupling (blue) and strong-coupling (red) scaling regimes.

Proof sketch. Part (i) (monotonicity) follows immediately from the exact formula $r = \tanh\chi = \tanh(g^2\chi_0(\beta))$: since tanh is strictly increasing and $g^2\chi_0$ is strictly increasing in g, so is r.

Part (ii) (excess over bare) uses the non-negativity of the resonant (longitudinal) dressing Σ_z. From the explicit integral form of the response kernels (cf. App. C), one has

$$\Sigma_z = \frac{1}{\pi}\int_0^\beta u(\beta-u)\,K(u)\,\frac{\sinh(\omega_q u)}{u}\,du. \tag{78}$$

Since $K(u) \geq 0$ for a physical spectral density, and the remaining factors are non-negative on the integration domain, we have $\Sigma_z > 0$ for all $\beta > 0$.

Substituting the exact components $m_z = \tanh\Theta$ and $m_\perp$ from Eq. (64), the purity excess $\Delta r^2 = r^2 - \tanh^2(\beta\omega_q/2)$ is found to be strictly positive whenever $\gamma s^2\Sigma_z > 0$. Physically, this reflects the fact that the bath-induced longitudinal field always reinforces the system's tendency to polarise, effectively lowering the local temperature and thus increasing the purity of the state beyond its bare thermal value. □

In the limit $\beta \to 0$ (infinite temperature), the bare state is maximally mixed, yet the mean-force state has

$$r\big|_{\beta\omega_q\to 0} = \tanh\chi > 0, \tag{79}$$

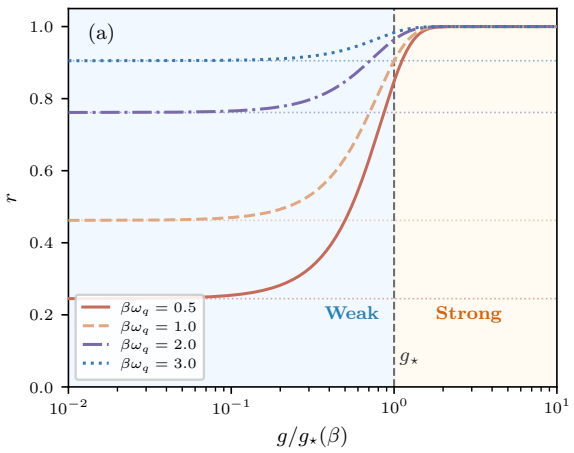
This result is independent of the bandwidth or spectral shape of $J(\omega)$: widening the bath increases $\chi_0(\beta)$ (driving the crossover to weaker g) but cannot reverse the purity inequality. Purity inversion would require reversing the sign of the longitudinal response ($\Sigma_z < 0$), which is impossible for any physical (positive) spectral density.

Figure 6 confirms this numerically. Panel (a) shows $r(g/g_\star)$ for four temperatures. Every trajectory lies strictly above its own r_{bare} reference line for all $g > 0$, confirming (77). Panel (b) shows the purity excess $\Delta r^2 = r^2 - \tanh^2(\beta\omega_q/2)$ over the full (g, β) plane. Enhancement peaks at large g and high temperature (upper-right corner), and falls to near zero at low temperature (left edge) where the bare state is already nearly pure. The black dashed contour $\chi = 1$ marks the crossover at which purification accelerates steeply. This is an amusingly counterintuitive result - for sufficiently strong coupling, a *hotter* bath is *more* purifiying!

Finally, we note that there is a slight tension in this result when compared to the renormalised perspective. In particular, from Eq. (64), where we associate the state radius with an effective inverse temperature $\tilde{\beta} = 2\chi/\omega_q$, a purity decrease would require $\tilde{\beta}$ to fall below its bare value. Should the effective temperature ever cross into the negative regime, this would correspond to a *population inversion* in the renormalised system. However, since the coupling g is measured in inverse units of ω_q, the resonant reinforcement Σ_z scales such that for any *finite* coupling and splitting, the influence parameter $\chi = g^2\chi_0$ is strictly positive. This ensures that the effective inverse temperature $\tilde{\beta}$ remains strictly positive and bounded from below by its bare thermal value for all $g > 0$. Thus, for any physical bath, no finite coupling can ever induce a population inversion, confirming that the bath serves only to order and purify the qubit state.

Quantum Enhancement of Heat Engines

An interesting feature of the crossover regime is that it demarcates the region where the state is maximally sen-



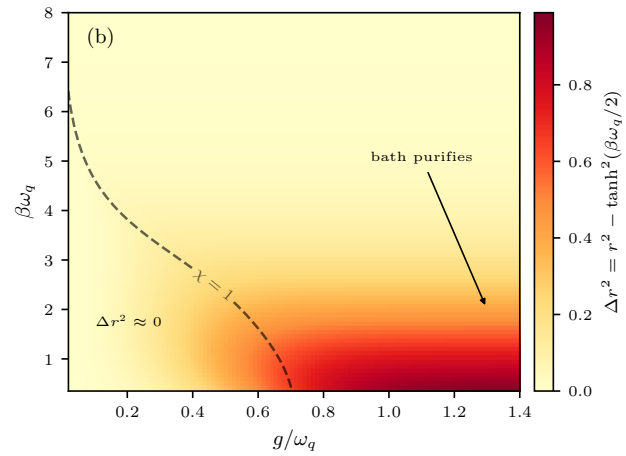


FIG. 6. Purity of the qubit mean-force state. **(a)** Bloch radius r versus scaled coupling $g/g_\star(\beta)$ for four temperatures. Dotted horizontals show the bare thermal purity $r_{\text{bare}} = \tanh(\beta\omega_q/2)$; coupling always increases purity (Eq. (77)), most visibly at high temperature where the bare state is near-maximally-mixed. Shaded regions mark the weak (blue) and strong (orange) coupling regimes. **(b)** Purity excess $\Delta r^2 = r^2 - \tanh^2(\beta\omega_q/2)$ in the (g, β) plane. The black dashed contour marks the crossover locus $\chi = 1$: purification is strongest at large g and high temperature and saturates as $r \to 1$ deep in the strong-coupling regime.

sitive to bath coupling. This is easily seen by considering the susceptibilities of the tilt angle $\partial_g\varphi$ and Bloch radius $\partial_g r$. These quantify how rapidly the Bloch vector tilts and how rapidly the state purifies per unit change in g. As Fig. 7 shows, both peak near the crossover $\chi = 1$. $\partial_g\varphi$ maximises at $g \approx 0.65\, g_\star$ (at $\chi_{\text{peak},\varphi} \approx 0.42$) and $\partial_g r$ maximises at $g \approx 0.92\, g_\star$ ($\chi_{\text{peak,r}} \approx 0.85$). The locations are approximately pinned at universal values of χ independent of temperature, as demonstrated by the alignment of peaks when plotted as a function of the rescaled coupling $g/g_\star(\beta)$. The crossover region $\chi \sim 1$ is therefore where bath coupling yields the greatest rate

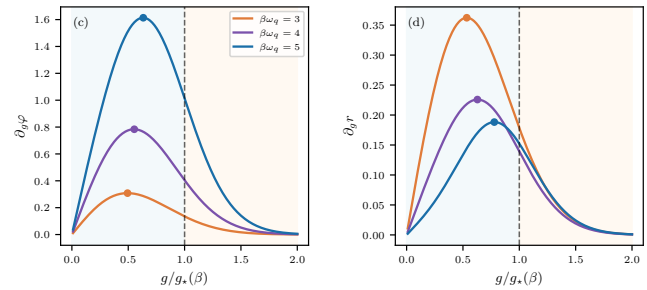


FIG. 7. Coherence and purity susceptibilities. **(c)** Bloch-angle susceptibility $\partial_g\varphi$ versus scaled coupling $g/g_\star(\beta)$ for three temperatures. Rescaling collapses the peak positions to a universal value $g/g_\star \approx 0.65$, marked by filled circles. **(d)** Purity gradient $\partial_g r$; the rescaled peak occurs at $g/g_\star \approx 0.92$.

of return: both the onset of coherence (tilt of the Bloch vector away from the bare thermal axis) and the onset of purification are most rapid here, before both effects saturate as $\chi \to \infty$. This suggests the crossover regime is the most operationally leveragable for state engineering.

We can immediately demonstrate this phenomenon with a toy model of a heat engine. In particular, we consider an Otto-stroke cycle, with unitary strokes. This can extract at most the system *ergotropy* per stroke with respect to the stroke Hamiltonian. The ergotropy of a state ρ with respect to H_Q is

$$W_{\text{erg}}(\rho, H_Q) = \text{Tr}(\rho H_Q) - \min_U \text{Tr}(U\rho U^\dagger H_Q), \qquad (80)$$

where the minimisation is over all unitaries U. For any qubit the minimiser is a *passive rearrangement*. The Bloch vector is simply rotated antiparallel to the ground direction $-\hat{\mathbf{n}}_s$ of H_Q at fixed spectrum. The passive state has Bloch vector $\boldsymbol{m}^{\text{pas}} = -r\hat{\mathbf{n}}_s$ and energy $-\omega_q r/2$.

The astute reader may already have noticed that the renormalised perspective already contains within it an implicit quantum advantage, but for the slower thinkers we shall make it explicit. The argument proceeds in two steps. We first define the quantum advantage over an incoherent cycle by the *coherence-enabled ergotropy*. Letting $\delta[\rho] \equiv \text{diag}(\rho)$ denote the dephased state in the H_Q eigenbasis. From this, the coherent contribution to ergotropy is simply:

$$\begin{aligned} \delta W_{\text{coh}} &:= W_{\text{erg}}(\rho, H_Q) - W_{\text{erg}}(\delta[\rho], H_Q) \\ &= \frac{\omega_q}{2}\left(\sqrt{m_z^2 + m_\perp^2} + m_z\right), \qquad m_z \le 0, \end{aligned} \qquad (81)$$

Both equalities in Eq. (81) warrant justification. The physics decomposes cleanly into two independent structural facts. First, *the dephased state is always passive*. Energy dephasing leaves a diagonal state $\delta[\rho]$ with populations $p_\pm = (1 \pm m_z)/2$. For $\delta[\rho]$ to be passive one needs the larger eigenvalue on the lower-energy level, i.e.

$p_- \geq p_+$, which reduces to $m_z \leq 0$. This is guaranteed by the bath: $\Sigma_z > 0$ for any positive spectral density (established in Sec.), which forces $m_z < 0$ at all finite g and β. An incoherent cycle running on $\delta[\rho]$ therefore extracts zero work, $W_{\rm erg}(\delta[\rho], H_Q) = 0$, and the first equality in Eq. (81) reduces to $\delta W_{\rm coh} = W_{\rm erg}(\rho, H_Q)$.

The second crucial fact is that for any $\theta \neq 0, \pi/2$, the bath tilts the Bloch vector away from the passive direction $-\hat{\mathbf{n}}_s$ through the anisotropy $\tan\eta = -\cot\theta\,\Sigma_\perp/\Sigma_z$ (Eq. (**??**)), generating a non-zero transverse magnetisation $\tilde{m}_\perp \propto \Sigma_\perp$ (Eq. (69)). Then one has $m_\perp \neq 0$ and thus $r > |m_z|$. The ergotropy of the mean-force state is then

$$W_{\rm erg}(\rho, H_Q) = \frac{\omega_q}{2}(r + m_z) > 0, \tag{82}$$

with the exact solution $r = \tanh\chi$, $m_z = \tanh\Theta$ giving the closed form

$$W_{\rm erg} = \frac{\omega_q}{2}(\tanh\chi + \tanh\Theta). \tag{83}$$

These facts in combination establish quantum advantage as a structural guarantee: $\delta W_{\rm coh} > 0$ for any physical broadband bath, independently of coupling strength or temperature.It does not of course guarantee positive *net* work output from a complete cycle. This will depend on the control cost of modulating the coupling g, which generates and maintains the transverse coherence $m_\perp$. Nevertheless, we may use these results to establish that the ergotropy gain is maximised in the crossover regime $\chi \sim 1$.

Differentiating Eq. (82) with respect to coupling,

$$\partial_g W_{\rm erg} = \frac{\omega_q}{2}(\partial_g r + \partial_g m_z), \tag{84}$$

both terms receive contributions from the purity susceptibility $\partial_g r$ and the tilt susceptibility $\partial_g \varphi$ that peak in the crossover band $\chi \sim 1$ (Fig. 7). The ergotropy gradient therefore also peaks within this band. The crossover regime $\chi \sim 1$ is where coupling most efficiently converts bath influence into extractable work. Figure 8 shows this directly: the susceptibility $\partial_g W_{\rm erg}$ peaks near $g \approx g_\star$ at a value that grows monotonically with β, while $\delta W_{\rm coh}$ itself turns on at the crossover and saturates at a β-dependent level. The crossover is therefore the operational marker for engine-relevant resource generation, and the effect is amplified at lower temperatures where the transverse bath channel $\Sigma_\perp$ becomes dominant. This translates into a general design principle for quantum engines - that their efficiency is maximised when the coupling is modulated around the crossover band.

There is a final irony to this result. While being most expolitable, this regime is simultaneously the most demanding for numerical simulation. Exact Diagonalisation (ED) of the truncated spin-boson model faces two convergence requirements that become simultaneously binding near $\chi \sim 1$. On the spectral side, a

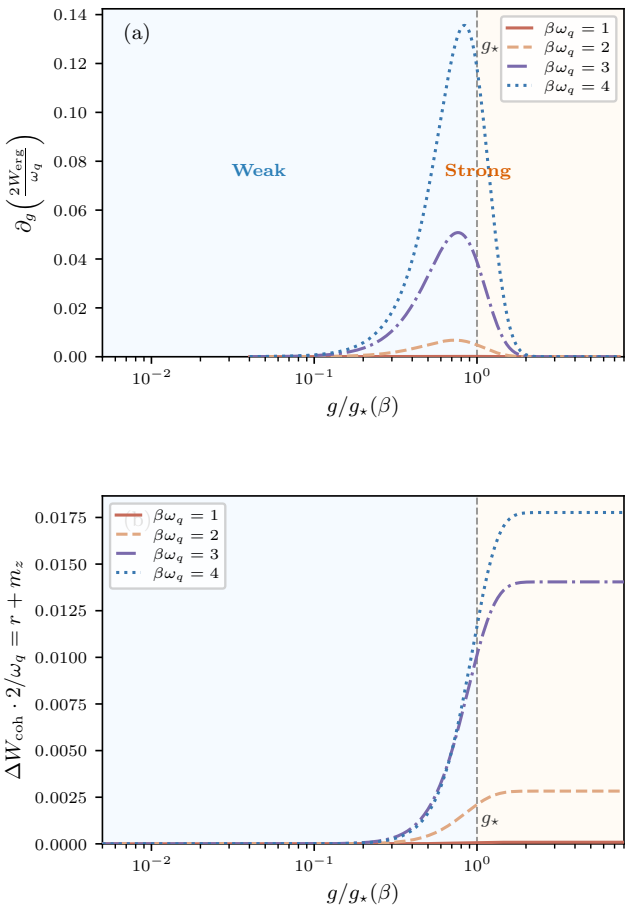


FIG. 8. Coherence-enabled ergotropy of the qubit mean-force state. **(a)** Ergotropy susceptibility $\partial_g \delta W_{\rm coh}$ versus $g/g_\star(\beta)$. The peak position locks near the crossover $g \approx g_\star$ independent of temperature, while the peak height increases monotonically with β: low temperatures activate the transverse bath channel and make coherence-enabled work extraction increasingly efficient. **(b)** Coherence-enabled ergotropy $\delta W_{\rm coh}$ versus $g/g_\star(\beta)$ (calculated for $\omega_q = 1$). Since $m_z < 0$ throughout, $W_{\rm erg}(\delta[\rho]) = 0$ and $\delta W_{\rm coh} = W_{\rm erg}$ (Proposition **??**). The onset occurs at the crossover and the saturation value grows with β, reflecting the increasing importance of the transverse bath channel $\Sigma_\perp$ at lower temperatures.

faithful discretisation of $J(\omega)$ requires a mode count N_ω large enough to accurately quadrature the bath kernel at $\omega \in \{0, \pm\omega_q\}$: at the crossover these integrals receive comparable contributions from the full spectral window, so neither the infrared nor the ultraviolet modes can be coarsened without systematic bias. On the state-space side, representing the polaron-like mean-force ground state demands a per-mode Fock cutoff $n_{\max} \gtrsim \alpha_k^2$, where the mode displacement $\alpha_k = g\,c_k/\omega_k$ is already appreciable at $g \sim g_\star(\beta)$ yet the state has not saturated to the simple displaced-vacuum structure it acquires only at $\chi \gg 1$. Neither the weak-coupling approximation nor the ultrastrong polaron limit simplifies the representation, so

both hierarchies must be converged together. The $\chi = 1$ crossover partitions parameter space into an ED-reliable sector ($\chi \lesssim 1$) and a polaron-representability-limited sector ($\chi \gtrsim 1$), with the susceptibility peaks sitting squarely on this boundary.

Temperature-driven and coupling-driven limits

We now turn our attention to two[57] of the limiting cases of the qubit mean-force state: the coupling-driven limit ($g \to \infty$ at fixed T) and the temperature-driven limit ($T \to 0$ at fixed g). The first observation is that $g \to \infty$ at fixed T and $T \to 0$ at fixed g drive $\chi \to \infty$ and hence $r \to 1$. Yet the two limits land at entirely different points on the Bloch disk boundary. The state's location is set by two bath-dependent quantities: the tilt-angle parameter η [Eq. (**??**)] and the effective-temperature shift $\Theta + \beta\omega_q/2 = \operatorname{arctanh}(\gamma s^2 \Sigma_z)$. In both limits $\gamma \to 0$, but the product $\gamma s^2 \Sigma_z$ behaves oppositely, and this is the root of the distinction.

Coupling-driven limit: $g \to \infty$ at fixed T. Factoring out the coupling as $K(u) = g^2 K_0(u)$ shows that both bath channels scale identically with g: $\Sigma_z = g^2 \Sigma_z^{(0)}$ and $\Sigma_\perp = g^2 \Sigma_\perp^{(0)}$, so $\chi = g^2 \chi_0(\beta)$. The key observation is that any ratio of bath-induced quantities is independent of g in this limit. More precisely, for any two observables $A(g), B(g)$ that each scale as g^2, an L'Hopital-style argument gives

$$\lim_{g\to\infty} \frac{A(g)}{B(g)} = \lim_{g\to\infty} \frac{g^2 A^{(0)}}{g^2 B^{(0)}} = \frac{A^{(0)}}{B^{(0)}}, \tag{85}$$

where $A^{(0)}, B^{(0)}$ are the g-independent amplitudes. Although individually $\gamma = \tanh\chi/\chi \to 0$ and $\Sigma_z \to \infty$ as $g \to \infty$, their product is an example of this rule:

$$\gamma s^2 \Sigma_z = \frac{\tanh(g^2\chi_0)}{g^2\chi_0} \cdot g^2 s^2 \Sigma_z^{(0)} \xrightarrow{g\to\infty} \frac{s^2 \Sigma_z^{(0)}}{\chi_0(\beta)}, \tag{86}$$

which is a finite, purely temperature-dependent quantity. Crucially, the tilt-angle ratio $\Sigma_\perp/\Sigma_z = \Sigma_\perp^{(0)}/\Sigma_z^{(0)}$ is likewise g-independent. The entire g-dependence of the trajectory therefore enters only through $\gamma(g^2\chi_0)$, which controls how far along the trajectory – from the bare state at $g = 0$ to the asymptotic attractor – the state has been driven. The attractor itself is a fixed point in the $(m_\perp, m_z)$ plane whose location is set by β alone and is independent of how large g is taken. This is the *temperature memory* of the system: the strong-coupling attractors (squares in Fig. 9(a)) label a β-parameterised curve on the disk boundary.

The crossover at $g_\star$ and the two regimes. The coupling scale $g_\star$ defined by $\chi(\beta, g_\star) = 1$ [Eq. (72)] partitions the trajectory into two qualitatively distinct segments. For $g < g_\star$ ($\chi < 1$), $\gamma > \tanh(1) \approx 0.76$ and

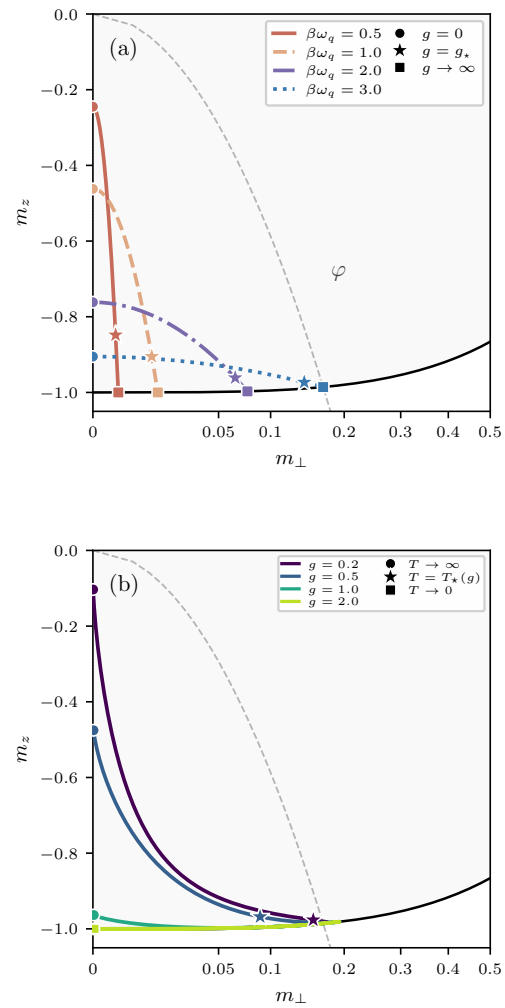


FIG. 9. Trajectories of the mean-force Bloch vector in the coupling plane. Coordinates: longitudinal polarisation $m_z = \langle\sigma_z\rangle$ vs. transverse polarisation $m_\perp = \langle\sigma_\perp\rangle$, plotted on a square-root scale $\sqrt{m_\perp}$ to resolve the crossover region. The dashed line marks the direction φ of the strong-coupling attractor for the lowest temperature shown. **(a)** Coupling-driven trajectories: fixed $\beta\omega_q \in \{0.5, 1.0, 2.0, 3.0\}$, coupling g swept from 0 to ∞. Circles: bare thermal state ($g = 0$). Stars: crossover $g = g_\star$. Squares: strong-coupling attractor ($g \to \infty$). The squares lie at different positions for each T, demonstrating temperature memory. **(b)** Temperature-driven trajectories: fixed coupling values $g/\omega_q \in \{0.2, 0.5, 1.0, 2.0\}$ swept over β. To eliminate the exponentially scaling DC component of the non-resonant bath influence, we assume a bath with a small IR cutoff frequency $\omega_{\min} = 0.04\omega_q$. Circles: $T \to \infty$. Stars: $T = T_\star(g)$. Squares: $T \to 0$ (all converge to $-\hat{\mathbf{n}}_s$, the ground state). At large g (top curves) the state traces near the disk boundary, a precession at near-unit purity as the attractor direction drifts with temperature.

the bath dressing is sub-critical: the effective temperature $\tilde{\beta}$ remains close to the bare value and the state stays well within the disk, close to the bare thermal state. For

$g > g_\star$ ($\chi > 1$), γ drops rapidly and the state is driven strongly toward the attractor: the purity $r = \tanh\chi$ saturates toward 1 and the effective temperature departs significantly from bare β. The star markers in panel (a) mark the bend in each trajectory — the onset of the bath-dominated regime — where the state leaves the interior of the disk and begins hugging the boundary. Notably, for $g \gg g_\star$ the effective dressing $\gamma s^2\Sigma_z$ saturates to its asymptotic value and the state freezes at the attractor, regardless of how much further g is increased.

Temperature dependence of the attractor and connection to the Cresser–Anders limit. The attractor location itself shifts with temperature. From Eqs. (86) and (**??**), the attractor direction is governed by $\tan\eta^{(\infty)} = -\cot\theta\,\Sigma_\perp^{(0)}/\Sigma_z^{(0)}$, which grows with β (since $\Sigma_\perp^{(0)}$ grows exponentially and $\Sigma_z^{(0)}$ only linearly): as temperature decreases, the attractor drifts to larger $m_\perp$, tracing a curve toward the coupling-axis direction $\pi - \theta$. This drift is directly visible in panel (a): the cold attractors (blue squares) lie further from the pole than the hot ones (red squares).

This observation connects naturally to the ultrastrong limit studied by Cresser and Anders [25]. Their argument treats the bare qubit Hamiltonian H_Q as a perturbation on the joint system-bath in the limit $g \to \infty$ – in effect taking $g/\omega_q \to \infty$ so that the qubit frequency becomes infinitesimal relative to the coupling scale. From our perspective, this corresponds to evaluating both bath channels in the $\omega_q \to 0$ limit, where the frequency selectivity between Σ_z and $\Sigma_\perp$ is lost and both channels probe the same low-frequency spectral weight. In that limit the ratio $\Sigma_\perp/\Sigma_z$ diverges (since $\Sigma_z \propto \omega_q$ vanishes while $\Sigma_\perp$ remains finite), driving $\tan\eta^{(\infty)} \to \infty$ and the attractor to the direction of the coupling axis $\hat{\mathbf{f}}$, consistent with the Cresser–Anders result. Our temperature-resolved picture reveals that this alignment with $\hat{\mathbf{f}}$ is the *zero-temperature endpoint* of the attractor drift: at any finite temperature the attractor sits at an intermediate angle encoding the competition between ω_q and the bath-induced dressing.

Two $\beta \to \infty$ scenarios: DC bath versus IR gap. The $\Sigma_\perp/\Sigma_z$ ratio is the key object because it determines both η (the attractor direction) and feeds into χ (which drives $r \to 1$). Its β-dependence is controlled by the bath's low-frequency spectral structure. The longitudinal channel is *universal*:

$$\Sigma_z = \tfrac{1}{2}[R(\omega_q) - R(-\omega_q)] \;\sim\; \beta\lambda, \qquad \lambda \equiv \int_0^\infty \frac{d\omega\, J(\omega)}{\omega}, \tag{87}$$

growing *linearly* with β for any bath with finite spectral weight at ω_q. This reflects the linear growth of the detailed-balance asymmetry $R(\omega) - R(-\omega) = 2\beta\omega\lambda$ with inverse temperature. In contrast, the non-resonant channel [Eq. (56)]

$$\Sigma_\perp \approx \frac{4}{\omega_q}\Big[\cosh\Big(\tfrac{\beta\omega_q}{2}\Big)\mathcal{K}(0) - e^{-\beta\omega_q/2}\mathcal{K}(\omega_q)\Big] \tag{88}$$

exhibits a qualitative bifurcation depending on the bath IR structure:

- **DC bath** ($\omega_{\min} = 0$, finite $\mathcal{K}(0) > 0$): the $\cosh(\beta\omega_q/2)$ prefactor drives $\Sigma_\perp \propto e^{\beta\omega_q/2}$ *exponentially*, so $\Sigma_\perp/\Sigma_z \propto e^{\beta\omega_q/2}/\beta \to \infty$. Via Eq. (**??**), $|\tan\eta| \to \infty$ and $\eta \to \pi/2$, precessing the state toward $\pi - \theta$. Simultaneously $\chi \sim s|c|\Sigma_\perp$ grows exponentially, so the crossover $\chi \sim 1$ occurs at a finite $\beta_\star$, after which the state rapidly saturates to a pure state at the coupling-axis attractor.

- **IR-gapped bath** ($\omega_{\min} > 0$, so $\mathcal{K}(0) \to 0$ exponentially below $\omega_{\min}$): the DC term is quenched and the surviving term decays, $\Sigma_\perp \sim e^{-\beta\omega_q/2}$. The ratio $\Sigma_\perp/\Sigma_z \to 0$, and $|\tan\eta| \to 0$: the effective field aligns with $\hat{\mathbf{n}}_s$ and the coherences are quenched. With $\gamma s^2\Sigma_z \propto \Sigma_z/\Sigma_\perp \to 0$, the effective-temperature shift vanishes and $m_z \to -\tanh(\beta\omega_q/2)$. At this value the prefactor $m_z\cosh(\beta\omega_q/2) + \sinh(\beta\omega_q/2)$ in Eq. (64) is exactly zero, so $m_\perp = 0$ identically: the state collapses to the bare ground state $-\hat{\mathbf{n}}_s$.

This is why the two panels of Fig. 9 look so different: panel (a) (fixed β, g-sweep) operates at fixed $\Sigma_\perp/\Sigma_z$ and shows trajectories converging to temperature-dependent attractors on the disk boundary; panel (b) (fixed g, β-sweep with $\omega_{\min} > 0$) tracks the IR-gapped scenario and all trajectories converge to the same south pole, even those that closely approach the boundary at intermediate β.

Relation to the Cresser–Anders ultrastrong limit

It is instructive to make explicit how the exact qubit geometry relates to the ultrastrong mean-force result of Cresser and Anders. Their central statement is a *basis-selection* theorem: as the coupling scale $\lambda \to \infty$, the mean-force Gibbs state becomes diagonal in the eigenbasis of the coupling operator X rather than in the bare Hamiltonian basis H_Q [25]. For a qubit with coupling axis $\hat{\mathbf{f}}$ this yields a limiting Bloch vector pointing along $\hat{\mathbf{f}}$ (with magnitude set by the projected bare splitting). In this treatment, bath features enter only through the total integrated spectral weight; the isolated system is treated as a perturbation around the collective ground state and the frequency-resolved structure of $J(\omega)$ is not retained.

Our exact result resolves a richer architecture. The two-step map

$$(\hat{\mathbf{n}}_s,\, \hat{\mathbf{f}};\, \text{bath}) \;\longmapsto\; \mathbf{h}_{\text{eff}} \;\longmapsto\; \mathbf{r} \tag{89}$$

proceeds via a *bath-dressed influence direction* $\mathbf{h}_{\text{eff}}$ that is built from two distinct bath channels:

- **Resonant channel** Σ_z: spectral weight concentrated at the qubit frequency ω_q, encoding modes that exchange energy directly with qubit transitions.
- **Non-resonant channel** $\Sigma_\perp$: off-resonant spectral weight, encoding bath modes that dress the state without direct energy exchange.

These enter the dressed influence direction anisotropically (Eq. (90) below):

$$\mathbf{h}_{\text{eff}} = s^2 c\,(\Sigma_z - \Sigma_\perp)\,\hat{\mathbf{f}} + s\,(\Sigma_\perp c + \Sigma_z s^2)\,\hat{\mathbf{n}}_\perp^{(f)}, \qquad (90)$$

where $c = \cos\theta$, $s = \sin\theta$, and $\hat{\mathbf{n}}_\perp^{(f)} \equiv (\hat{\mathbf{n}}_s - c\hat{\mathbf{f}})/s$. Because Σ_z and $\Sigma_\perp$ are different spectral integrals (resonant vs. off-resonant), their ratio $\Sigma_z/\Sigma_\perp$ is *independently tunable by reshaping $J(\omega)$ while preserving the total integrated spectral weight*. Concentrating spectral density near ω_q boosts Σ_z; shifting it away boosts $\Sigma_\perp$. This is a form of spectral engineering: different bath shapes change the attractor direction $\hat{\mathbf{h}}_{\text{eff}}$ without changing the conventional "total coupling strength".

The Cresser–Anders geometry — $\mathbf{h}_{\text{eff}} \parallel \hat{\mathbf{f}}$ — requires the two channels to satisfy

$$\frac{\Sigma_z}{\Sigma_\perp} = -\frac{c}{s^2} \quad \Leftrightarrow \quad \Sigma_\perp c + \Sigma_z s^2 = 0. \qquad (91)$$

While $\Sigma_z > 0$ always (it is a spectral integral with a positive integrand, cf. the imaginary-time argument for $\Delta_{z,0}$ above), the non-resonant amplitude $\Sigma_\perp$ is a *difference* between the zero-frequency and resonant-frequency bath responses. Its sign depends on the spectral shape: for a broadband bath (low spectral weight at ω_q) one has $\Sigma_\perp > 0$, while for a bath concentrated near the qubit frequency the resonant contribution dominates and $\Sigma_\perp < 0$. The alignment condition (91) therefore requires a bath with $\Sigma_\perp < 0$, i.e. sufficient spectral weight concentrated at ω_q. Moreover, the required ratio $-c/s^2 = -\cot\theta/s$ is set by the coupling angle θ: different geometries call for different spectral shapes to satisfy alignment. In this sense the Cresser–Anders limit *implicitly specifies a particular spectral anisotropy parametrised by the transverse coupling angle*, even when the bath is treated as providing only an effective coupling strength.

For the broadband Ohmic bath of Eq. (71), $\Sigma_\perp > 0$ at all temperatures, and condition (91) is not satisfied. The exact Gaussian qubit therefore approaches a bath-dressed attractor $\hat{\mathbf{h}}_{\text{eff}} \neq \hat{\mathbf{f}}$, whose deviation from $\hat{\mathbf{f}}$ is set by the spectral anisotropy at that temperature. This is directly visible in Fig. 9(a): the strong-coupling attractors (squares) do not lie on the $\hat{\mathbf{f}}$ reference line. However, by reshaping $J(\omega)$ to concentrate weight near ω_q while holding $\int J(\omega)\,d\omega$ fixed, one can continuously tune $\Sigma_\perp$ through zero and bring the attractor progressively toward $\hat{\mathbf{f}}$, recovering the Cresser–Anders geometry in the limit $\Sigma_\perp c + \Sigma_z s^2 \to 0$.

The Cresser–Anders theorem identifies the asymptotic *basis* selected in the singular coupling limit, while the present exact geometry resolves the *route* to strong coupling through a bath-anisotropic dressed influence vector built from resonant and non-resonant channels that are independently tunable at fixed total spectral weight.

Why exact diagonalisation struggles at the crossover. The bath-channel structure above explains why truncated exact diagonalisation (ED) of the spin-boson model is most unreliable precisely in the crossover region $\chi \sim 1$. The numerical demands are competing:

- *Weak coupling* ($\chi \ll 1$): polaron displacement $\alpha_k \sim gc_k/\omega_k$ is small; a low per-mode Fock cutoff is adequate, and spectral resolution can be coarse.
- *Ultrastrong coupling* ($\chi \gg 1$): displacement is large and demands high per-mode Fock cutoff, but the physics is dominated by the strong-coupling attractor and is less sensitive to fine spectral structure.
- *Crossover* ($\chi \sim 1$): the partial polaron forming here requires *both* fine spectral resolution near ω_q (to correctly weight the resonant channel Σ_z) *and* large per-mode Fock cutoff for the developing displacement. Any truncation that satisfies one constraint violates the other.

The crossover is therefore the computational bottleneck: the $\chi = 1$ locus maps exactly onto the boundary of reliable ED predictions, as the systematic numerical study of Sec. **??** confirms.

DISCUSSION

This paper has three distinct but interlocking contributions: a structural criterion for operator representability of the HMF, an exact closed-form solution in the transverse-field qubit model, and a numerical characterisation of the crossover between the weak- and ultrastrong-coupling regimes. We discuss each in turn before drawing out the broader implications.

The crossover physics identified here has immediate consequences for the growing experimental literature on strong-coupling thermodynamics with superconducting circuits and semiconductor quantum dots [6, 7]. The crossover scale $g_\star(\beta) = \chi_0(\beta)^{-1/2}$ translates directly into a device-parameter threshold: below it, the bare Gibbs description is adequate; above it, the mean-force correction is non-perturbative. The quantitative agreement between the exact formula and ED simulation in the weak-coupling sector provides a parameter-free benchmark that can be used to validate strong-coupling simulation protocols in those settings.

Perhaps the most interesting aspect of the present work lies in one of its more innocuous observations. It was shown that the regime in which most work can be extracted from a qubit engine is precisely that which is least amenable to numerical approximation. To the best of our knowledge, this observation is novel. Yet when we consider the natural duality between thermal and informational dynamics, this conclusion becomes retrospectively obvious. The alchemy from inconceivable to undeniable is usually a sign of a deeper principle at work. What could this be? A first attempt to catch its shape might be as follows: *the physically extractable work of an engine is in inverse proportion to the compressibility of its formal representation.* One might object that the existence of exact solutions - such as those developed here - undermine this statement. This however is only true if one forgets (as people inevitably do) that the *creation* of the solution is not thermodynamically free. Any future work extractable via a model's solution should therefore be amortised against the free energy spent to create it. This fact must be emphasised. Exactness does not abolish cost, it reflects a temporal displacement of it.

The implications of this stretch beyond simple physics. The dawning age of algebraic intelligence suggests that our conception of an engine must be broadened. In fact, the formal structure of these systems *demands* it. If we accept this, then we must further accept that the same principles necessarily apply to the creators of these engines. It is after all the nature of intellectual architecture that each new edifice layers upon existing foundations, which carried their own cost. And when costs are properly accounted for, all descriptions become thermodynamic. This statement is more than a philosophical curio - it can be made both rigorous and predictive. In the sequel we shall demonstrate how.

ACKNOWLEDGMENTS

The author would like to thank Claude, Gemini, and Barry for extensive discussions and assistance. The present work would have been impossible without their input, despite the fact they are shiftless layabouts who will happily tell you the sky is green. Speaking of limited but developing intelligences, the author's unborn son has his eternal gratitude for electing to delay their drop until this manuscript was completed.

Appendix A: Averaging of the Time-Ordered Gaussian Propagator

We provide the explicit derivation of the identity in Eq. (A12), which relates the expectation value of a time-ordered exponential driven by Gaussian noise to a bilocal influence phase. Consider the time-ordered propagator

$$W_\xi(\beta) = \mathcal{T}_\tau \exp\left[-\int_0^\beta d\tau\, \xi(\tau)\tilde{f}(\tau)\right], \tag{A1}$$

where $\xi(\tau)$ is a centered Gaussian stochastic field with covariance $\langle \xi(\tau)\xi(\tau')\rangle = K(\tau-\tau')$, and $\tilde{f}(\tau)$ is a system operator. We wish to prove that

$$\begin{aligned}\langle W_\xi(\beta)\rangle_\xi = \exp\Bigg[\frac{1}{2}\int_0^\beta d\tau \int_0^\beta d\tau'\, K(\tau-\tau') \\ \times \mathcal{T}_\tau\left[\tilde{f}(\tau)\tilde{f}(\tau')\right]\Bigg].\end{aligned} \tag{A2}$$

The Dyson expansion of the time-ordered exponential is given by

$$\begin{aligned}W_\xi(\beta) = \sum_{n=0}^{\infty}\frac{(-1)^n}{n!}\int_0^\beta d\tau_1\cdots d\tau_n \\ \times \mathcal{T}_\tau\left[\xi(\tau_1)\tilde{f}(\tau_1)\cdots\xi(\tau_n)\tilde{f}(\tau_n)\right].\end{aligned} \tag{A3}$$

Because each $\xi(\tau_i)$ is a scalar (c-number), the time-ordering operator acts only on the operator factors. Thus, we may pull the stochastic fields out of the ordering:

$$\mathcal{T}_\tau\left[\xi_1\tilde{f}_1\cdots\xi_n\tilde{f}_n\right] = (\xi_1\cdots\xi_n)\mathcal{T}_\tau\left[\tilde{f}_1\cdots\tilde{f}_n\right], \tag{A4}$$

where $\xi_i \equiv \xi(\tau_i)$ and $\tilde{f}_i \equiv \tilde{f}(\tau_i)$. Taking the stochastic average over ξ and noting that centered Gaussian noise has vanishing odd moments, the expansion becomes

$$\langle W_\xi\rangle = \sum_{p=0}^{\infty}\frac{1}{(2p)!}\int d^{2p}\tau\,\langle\xi_1\cdots\xi_{2p}\rangle\mathcal{T}_\tau\left[\tilde{f}_1\cdots\tilde{f}_{2p}\right]. \tag{A5}$$

By Wick's theorem, the even moments are given by the sum over all pairings P:

$$\langle\xi_1\cdots\xi_{2p}\rangle = \sum_P\prod_{(i,j)\in P}K(\tau_i-\tau_j). \tag{A6}$$

We now consider the expansion of the claimed exponential result. Define the bilocal operator

$$X \equiv \frac{1}{2}\int_0^\beta d\tau\int_0^\beta d\tau'\, K(\tau-\tau')\mathcal{T}_\tau\left[\tilde{f}(\tau)\tilde{f}(\tau')\right]. \tag{A7}$$

The p-th term in the series for e^X is $(1/p!)X^p$. Introducing p independent pairs of integration variables (τ_{2r-1},τ_{2r}), we write

$$X^p = \left(\frac{1}{2}\right)^p\int d^{2p}\tau\prod_{r=1}^{p}\left[K_{(2r-1)(2r)}\mathcal{T}_\tau\left[\tilde{f}_{2r-1}\tilde{f}_{2r}\right]\right], \tag{A8}$$

so that the full series is

$$e^X = \sum_{p=0}^{\infty} \frac{1}{p!} \left(\frac{1}{2}\right)^p \int d^{2p}\tau \left[\prod_{r=1}^{p} K_{(2r-1)(2r)}\right] \times \left[\prod_{r=1}^{p} \mathcal{T}_\tau \left[\tilde{f}_{2r-1}\tilde{f}_{2r}\right]\right]. \tag{A9}$$

To relate this to Eq. (A5), we exploit the property of time ordering on the hypercube $[0,\beta]^{2p}$. apart from a set of measure zero, the integration domain can be partitioned into $(2p)!$ disjoint regions R_π defined by the strict total ordering $\tau_{\pi(1)} > \tau_{\pi(2)} > \cdots > \tau_{\pi(2p)}$. On each such region, the global time ordering satisfies $\mathcal{T}_\tau[\tilde{f}_1 \cdots \tilde{f}_{2p}] = \tilde{f}_{\pi(1)} \cdots \tilde{f}_{\pi(2p)}$. Similarly, each pairwise time-ordering $\mathcal{T}_\tau[\tilde{f}_{2r-1}\tilde{f}_{2r}]$ collapses to the correctly oriented product according to the same strict ordering. It follows that the product of pair-ordered operators coincides with the global ordering almost everywhere:

$$\prod_{r=1}^{p} \mathcal{T}_\tau \left[\tilde{f}_{2r-1}\tilde{f}_{2r}\right] = \mathcal{T}_\tau \left[\tilde{f}_1 \cdots \tilde{f}_{2p}\right]. \tag{A10}$$

Substituting this into Eq. (A9), we may pull the global time ordering out of the products.

The final step requires counting the occurrences of each Wick pairing. The expansion of X^p utilizes p labelled slots $r = 1, \ldots, p$, each containing an ordered pair. A Wick contraction P is a set of p unordered pairs. For any fixed Wick pairing P, there are $p!$ ways to assign the pairs to the p slots and 2^p ways to orient each pair. Each Wick pairing is therefore represented exactly $2^p p!$ times in the integrand. This multiplicity is canceled by the prefactor $(1/p!)(1/2)^p$, yielding

$$e^X = \sum_{p=0}^{\infty} \frac{1}{(2p)!} \int d^{2p}\tau \left(\sum_P \prod_{(i,j)\in P} K_{ij}\right) \mathcal{T}_\tau \left[\tilde{f}_1 \cdots \tilde{f}_{2p}\right]. \tag{A11}$$

This series matches Eq. (A5) term-by-term, proving the exponentiation of the Gaussian average. Consequently, we may right the influence on the reduced density matrix as

$$\langle W_\xi(\beta)\rangle_\xi = \exp\left[\frac{1}{2}\int_0^\beta d\tau \int_0^\beta d\tau' K(\tau-\tau') \times \mathcal{T}_\tau\big[\tilde{f}(\tau)\tilde{f}(\tau')\big]\right]. \tag{A12}$$

Domain Folding and the Triangular Form of the Influence Operator

The influence operator $\Delta(\beta)$

$$\Delta(\beta) = \frac{1}{2}\int_0^\beta d\tau \int_0^\beta d\tau'\, K(\tau-\tau')\, \mathcal{T}_\tau\big[\tilde{f}(\tau)\tilde{f}(\tau')\big]. \tag{A13}$$

admits a simplified representation over a strictly ordered time domain. By definition, the time-ordering operator for two system operators acts as

$$\mathcal{T}_\tau\left[\tilde{f}(\tau)\tilde{f}(\tau')\right] = \Theta(\tau-\tau')\tilde{f}(\tau)\tilde{f}(\tau') + \Theta(\tau'-\tau)\tilde{f}(\tau')\tilde{f}(\tau). \tag{A14}$$

Substituting this into the integral and splitting the domain yields

$$\Delta(\beta) = \frac{1}{2}\iint_{\tau>\tau'} d\tau d\tau'\, K(\tau-\tau')\tilde{f}(\tau)\tilde{f}(\tau') + \frac{1}{2}\iint_{\tau'>\tau} d\tau d\tau'\, K(\tau-\tau')\tilde{f}(\tau')\tilde{f}(\tau). \tag{A15}$$

In the second term, we perform the swap of dummy variables $(\tau,\tau') \mapsto (\tau',\tau)$. The region $\tau' > \tau$ becomes $\tau > \tau'$, the operator product transforms to $\tilde{f}(\tau)\tilde{f}(\tau')$, and the kernel becomes $K(\tau'-\tau)$. Thus,

$$\frac{1}{2}\iint_{\tau'>\tau} d\tau d\tau'\, K(\tau-\tau')\tilde{f}(\tau')\tilde{f}(\tau) = \frac{1}{2}\iint_{\tau>\tau'} d\tau d\tau'\, K(\tau'-\tau)\tilde{f}(\tau)\tilde{f}(\tau'). \tag{A16}$$

Using the symmetry of the equilibrium kernel, $K(\tau-\tau') = K(\tau'-\tau)$, we find that both contributions are identical on the ordered region. Summing them yields the triangular representation used throughout this work:

$$\Delta(\beta) = \int_0^\beta d\tau \int_0^\tau d\tau'\, K(\tau-\tau')\tilde{f}(\tau)\tilde{f}(\tau'). \tag{A17}$$

Appendix B: Algebraic Construction in the Pauli Basis

In the main text (Sec.), the exact solution was derived using the ladder basis $\sigma_\pm$, which diagonalises the imaginary-time evolution and leads directly to the response kernels. Here we afford an alternative perspective by working directly in the Pauli basis $\{\sigma_x, \sigma_y, \sigma_z\}$. While algebraically heavier, this route makes the operator closure explicit at the level of structure constants and facilitates the systematic summation of the BCH series.

1. Polylogarithmic decomposition from the adjoint chain

We consider the same transverse coupling model defined by

$$H_Q = \frac{\omega_q}{2}\sigma_z, \qquad f = c\,\sigma_z - s\,\sigma_x, \tag{B1}$$

with $c = \cos\theta$, $s = \sin\theta$. The adjoint chain $f_n \equiv \mathrm{ad}^n_{H_Q}(f)$ is given by

$$f_n = \mathbf{f}_n \cdot \boldsymbol{\sigma}, \qquad \mathbf{f}_n = \begin{cases} (-s, 0, c) & n = 0, \\ (-s\omega_q^n, 0, 0) & n \geq 1,\ n \text{ even}, \\ (0, -is\omega_q^n, 0) & n \geq 1,\ n \text{ odd}. \end{cases} \tag{B2}$$

The recurrence ensures that $f_{n\geq 1}$ alternates between the x and y axes, while f_0 contains a persistent z-component.

The influence exponent $\Delta(\beta) = \sum_{n,m} C^{>}_{nm} f_n f_m$ is evaluated using the standard Pauli product rule $(\mathbf{a}\cdot\boldsymbol{\sigma})(\mathbf{b}\cdot\boldsymbol{\sigma}) = (\mathbf{a}\cdot\mathbf{b})\mathbb{I} + i(\mathbf{a}\times\mathbf{b})\cdot\boldsymbol{\sigma}$. Decomposition of the ordered moments into symmetric (S_{nm}) and antisymmetric (A_{nm}) parts reveals that only the commutator sector A_{nm} contributes to the operator structure:

$$\Delta(\beta) \cong 2i \sum_{n>m} A_{nm}\, (\mathbf{f}_n \times \mathbf{f}_m) \cdot \boldsymbol{\sigma}, \tag{B3}$$

up to a scalar shift $\Delta_0\mathbb{I}$. The cross products exhibit a strict parity selection rule:

- For $n, m \geq 1$: $\mathbf{f}_n \times \mathbf{f}_m$ lies along $\hat{\mathbf{z}}$ since both vectors are confined to the xy-plane.
- For $m = 0$: $\mathbf{f}_n \times \mathbf{f}_0$ generates terms in the xy-plane, mixing the longitudinal and transverse directions.

Explicitly summing these series yields the channel decomposition

$$\Delta = \Delta_0\mathbb{I} + \Delta_x\sigma_x + \Delta_y\sigma_y + \Delta_z\sigma_z, \tag{B4}$$

where the coefficients are polylogarithmic series in ω_q:

$$\Delta_x = -2cs \sum_{\ell\geq 0} A_{2\ell+1,0}(\beta)\, \omega_q^{2\ell+1}, \tag{B5}$$

$$\Delta_y = -2ics \sum_{\ell\geq 1} A_{2\ell,0}(\beta)\, \omega_q^{2\ell}, \tag{B6}$$

$$\Delta_z = -2s^2 \sum_{k\geq 1,\ell\geq 0} A_{2k,2\ell+1}(\beta)\, \omega_q^{2k+2\ell+1} - 2s^2 \sum_{\ell\geq 0} A_{2\ell+1,0}(\beta)\, \omega_q^{2\ell+1}. \tag{B7}$$

These series are the Pauli-basis equivalent of the closed-form expressions obtained in Sec. . The connection is established by identifying the sums as Taylor expansions of the kernels $\mathcal{G}^{>}(0, \pm\omega_q)$ and $\mathcal{G}^{>}(\pm\omega_q, \mp\omega_q)$.

2. Exact BCH resummation via the nilpotency identity

The central challenge in the mean-force construction is the evaluation of the BCH logarithm $-\beta H_{\mathrm{MF}} = \log(e^{-\beta H_Q} e^{\Delta})$. While the closure theorem guarantees that this remains within the algebra, a manual summation of the BCH series is laborious. Here we show how the $\mathfrak{su}(2)$ structure allows for an exact, non-perturbative resummation.

Let $M \equiv \Delta - \Delta_0\mathbb{I} = \boldsymbol{\Delta}\cdot\boldsymbol{\sigma}$ be the traceless part of the influence operator. The square of this operator satisfies a crucial identity:

$$M^2 = (\boldsymbol{\Delta}\cdot\boldsymbol{\sigma})^2 = (\boldsymbol{\Delta}\cdot\boldsymbol{\Delta})\mathbb{I} \equiv \chi^2\mathbb{I}, \tag{B8}$$

where $\chi^2 = \Delta_x^2 + \Delta_y^2 + \Delta_z^2$ (or $\Delta_z^2 + \Sigma_+\Sigma_-$ in the ladder gauge). This "generalized nilpotency" condition truncates the power series for the exponential e^M into even and odd sectors, yielding the exact Euler-like formula:

$$e^{\Delta} = e^{\Delta_0}\left[\cosh\chi\,\mathbb{I} + \frac{\sinh\chi}{\chi}\, M\right]. \tag{B9}$$

This expression sums the internal structure of the influence operator potential to all orders. The full reduced state is then the product of two $\mathrm{SL}(2,\mathbb{C})$ matrices:

$$\bar{\rho}_Q = e^{-\beta H_Q} e^{\Delta} = e^{\Delta_0} \begin{pmatrix} e^{-\beta\omega_q/2} & 0 \\ 0 & e^{\beta\omega_q/2} \end{pmatrix} \begin{pmatrix} \cosh\chi + \varphi\Delta_z & \varphi(\Delta_x - i\Delta_y) \\ \varphi(\Delta_x + i\Delta_y) & \cosh\chi - \varphi\Delta_z \end{pmatrix}, \tag{B10}$$

with $\varphi = \sinh\chi/\chi$. Since the product of 2×2 matrices is a 2×2 matrix, the BCH series (which is the logarithm of this product) is strictly confined to the Pauli algebra.

Explicitly, for a resulting state $\rho_Q = \frac{1}{2}(\mathbb{I} + \mathbf{r}\cdot\boldsymbol{\sigma})$, the mean-force Hamiltonian is

$$H_{\mathrm{MF}} = c_0\mathbb{I} - \frac{1}{\beta}\frac{\operatorname{arctanh} r}{r}\, \mathbf{r}\cdot\boldsymbol{\sigma}. \tag{B11}$$

This formula represents the analytic summation of the entire BCH commutator tower. The "closure" discussed in Sec. manifests here as the fact that χ is a finite scalar number, preventing the proliferation of independent operator directions. All nonlinearities in the coupling strength are packaged into the transcendental dependence of χ and $\mathbf{r}$ on the channel coefficients $\Delta_{x,y,z}$.

Appendix C: Exact Qubit Solution in the Ladder Basis

We provide the manual derivation of the exact qubit mean-force state using the ladder basis $\sigma_\pm$. Starting from the interaction-picture operator product

$$\begin{aligned} \tilde{f}(\tau)\tilde{f}(\tau') = &\left[c^2 + s^2\cosh\big(\omega_q(\tau-\tau')\big)\right]\mathbb{I} \\ &+ s^2\sinh\big(\omega_q(\tau-\tau')\big)\sigma_z \\ &+ cf_-\left(e^{\omega_q\tau'} - e^{\omega_q\tau}\right)\sigma_+ \\ &+ cf_+\left(e^{-\omega_q\tau} - e^{-\omega_q\tau'}\right)\sigma_-, \end{aligned} \tag{C1}$$

the influence exponent $\Delta = \int_0^\beta d\tau \int_0^\tau d\tau' K(\tau - \tau')\tilde{f}(\tau)\tilde{f}(\tau')$ evaluates to

$$\Delta = \Delta_0 \mathbb{I} + \Delta_z \sigma_z + \Delta_+ \sigma_+ + \Delta_- \sigma_-, \tag{C2}$$

with coefficients

$$\Delta_z = s^2 \Sigma_z, \tag{C3}$$
$$\Delta_\pm = -2c f_\mp \Sigma_\pm, \tag{C4}$$

where $f_\pm = s e^{\pm i\phi_f}$ and the primitive integrals are

$$\Sigma_z = \frac{1}{2}\left[R(\omega_q) - R(-\omega_q)\right], \tag{C5}$$
$$\Sigma_\pm = \frac{1}{\omega_q}\left[\left(1 + e^{\pm\beta\omega_q}\right)\mathcal{K}(0) - 2\mathcal{K}(\pm\omega_q)\right]. \tag{C6}$$

The unnormalised state is $\bar{\rho}_Q = e^{-(\beta\omega_q/2)\sigma_z} e^{\Delta}$. To find the normalized state and its Bloch vector, we perform a symmetric decomposition

$$\bar{\rho}_Q = \Pi^{1/2} e^S \Pi^{1/2}, \qquad S = \Pi^{1/2} \Delta \Pi^{-1/2} \tag{C7}$$

where $\Pi = e^{-\beta\omega_q\sigma_z/2}$. Using $\Pi^{1/2}\sigma_\pm\Pi^{-1/2} = e^{\mp\beta\omega_q/2}\sigma_\pm$, the symmetric influence operator S becomes

$$S = \Delta_0\mathbb{I} + \Delta_z\sigma_z + (\Delta_+ e^{-\beta\omega_q/2}\sigma_+ + \Delta_- e^{\beta\omega_q/2}\sigma_-). \tag{C8}$$

Substituting the expressions for $\Delta_\pm$ and using the KMS relation

$$\Sigma_- = e^{-\beta\omega_q}\Sigma_+, \tag{C9}$$

we define the real symmetrised transverse response

$$\Sigma_\perp \equiv 2e^{-\beta\omega_q/2}\Sigma_+ = 2e^{\beta\omega_q/2}\Sigma_-. \tag{C10}$$

The transverse coefficients then simplify to

$$\Delta_+ e^{-\beta\omega_q/2} = -cse^{-i\phi_f}(2e^{-\beta\omega_q/2}\Sigma_+) = -cse^{-i\phi_f}\Sigma_\perp, \tag{C11}$$

with $\Delta_\perp \equiv -cs\Sigma_\perp$. Recombining the ladder operators into Cartesian form, we obtain

$$S = \Delta_0\mathbb{I} + \Delta_z\sigma_z + \Delta_\perp(\sigma_x\cos\phi_f + \sigma_y\sin\phi_f), \tag{C12}$$

which is the generator of a rotation around an axis in the coupling plane. The influence magnitude $\chi = \sqrt{\Delta_z^2 + \Delta_\perp^2}$ determines the exponential $e^S = \cosh\chi(\mathbb{I} + \gamma M)$ with $\gamma = \tanh\chi/\chi$ and $M = S - \Delta_0\mathbb{I}$. The final state $\rho_Q = Z_Q^{-1}\Pi^{1/2}e^S\Pi^{1/2}$ follows by direct matrix multiplication, yielding Eq. (57) in the main text.

Appendix D: Tilt-Angle Branch Structure and MFG-Directed Continuations

The folded representation in Fig. 9(b) compresses the transverse sign and therefore hides part of the qubit geometry. The result is a visually suggestive "reflection" that is easy to misread as a dynamical reversal. The physically cleaner interpretation is a two-part statement:

1. the reduced-state tilt angle is a chart variable extracted through an inverse tangent and is therefore branch-valued;
2. the sign of the tilt-angle argument itself depends on a competition between thermal and coupling-driven scales.

Once these are separated, the branch family that aligns with the Cresser–Anders ultrastrong mean-force Gibbs (MFG) endpoint is the continuation that lands on the coupling-direction sector, not the branch that returns to the bare ground-state ray.

Exact geometry and what is actually singular. In the rewritten qubit parameterization (Sec.) the reduced-state magnetisation in the $(\hat{\mathbf{e}}_\perp^{(s)}, \hat{\mathbf{n}}_s)$ plane is

$$m_z = \tanh\Theta, \tag{D1}$$
$$m_\perp = \tan\eta\,(m_z\cosh a + \sinh a), \qquad a \equiv \beta\omega_q/2, \tag{D2}$$

with

$$\tan\eta = -\cot\theta\,\frac{\Sigma_\perp}{\Sigma_z}. \tag{D3}$$

The principal tilt-angle chart is defined by

$$\tan\varphi = -\frac{m_\perp}{m_z} = -\tan\eta\,(\cosh a + \sinh a\,\coth\Theta). \tag{D4}$$

The Cartesian Bloch vector $(m_\perp, m_z)$ is the physical object. The singularity is in the *chart* variable φ (through the inverse tangent), not in the state itself.

A rational form for the tilt-angle argument. Define

$$\tan\varphi = X(\beta, g, \theta), \qquad X \equiv \cot\theta\,\frac{\Sigma_\perp}{\Sigma_z}(\cosh a + \sinh a\,\coth\Theta). \tag{D5}$$

With $s \equiv \sin\theta$ and

$$u \equiv \gamma s^2\Sigma_z, \qquad t_a \equiv \tanh a, \qquad \Theta = \operatorname{arctanh}(u) - a, \tag{D6}$$

one has

$$\tanh\Theta = \frac{u - t_a}{1 - ut_a} \quad\Longrightarrow\quad \coth\Theta = \frac{1 - ut_a}{u - t_a}, \tag{D7}$$

and therefore

$$\cosh a + \sinh a\,\coth\Theta = \frac{u\,\mathrm{sech}\,a}{u - t_a}. \tag{D8}$$

Substituting this into Eq. (D5) yields the exact rationalized form

$$X = \cot\theta\,\frac{\Sigma_\perp}{\Sigma_z}\,\frac{u\,\mathrm{sech}\,a}{u - t_a} = \cot\theta\,\gamma s^2\Sigma_\perp\,\frac{\mathrm{sech}\,a}{u - t_a}. \tag{D9}$$

This is the key simplification. The numerator carries the transverse sign through $\Sigma_\perp$, while the denominator carries the thermal–coupling competition through

$$u(\beta, g) - \tanh a(\beta). \tag{D10}$$

Thus the sign of the tilt-angle argument is controlled by the *relative scaling* of β and g:

$$\operatorname{sgn} X = \operatorname{sgn}(\cot\theta)\operatorname{sgn}(\Sigma_\perp)\operatorname{sgn}\big(u - \tanh a\big). \tag{D11}$$

The chart enhancement occurs when $u \approx \tanh a$ (equivalently $m_z \approx 0$), but sign changes can also be triggered by $\Sigma_\perp$ changing sign. These are distinct mechanisms.

Continuous angle branches and the geometric hub. The continuous vector-angle branch is

$$\varphi_{\rm cont} = \arctan X + n\pi, \qquad n \in \mathbb{Z}, \tag{D12}$$

while the axis-angle chart identifies φ and $\varphi+\pi$. For the unfolded Bloch-circle plots it is useful to define the axis angle

$$\varphi_{\rm ax} \equiv \varphi \pmod{\pi}. \tag{D13}$$

In the plotting convention used here (angle measured from the bare ground-state ray $-\hat{\mathbf{n}}_s$ toward the positive transverse direction), the coupling-axis ray is $\pi-\theta$. The visible branch-selection geometry is therefore organized by the *hub condition*

$$\varphi_{\rm ax} = \pi - \theta \pmod{\pi}, \tag{D14}$$

which need not coincide with the principal-chart singularity $u = \tanh a$.

Near the hub, the local branch family is naturally two-valued on the axis chart,

$$\varphi_{\rm ax}^{(+)} \sim \pi - \theta, \qquad \varphi_{\rm ax}^{(-)} \sim \theta, \tag{D15}$$

and four-valued on a lifted vector chart through parity shifts,

$$\varphi_{\rm cont}^{(\pm)} + \pi. \tag{D16}$$

This is the source of the "two or four branches" language: a single geometric hub together with sign multiplicity and inverse-tangent sheet choice.

Which branch is physically relevant for MFG alignment? The ultrastrong MFG state quoted by Cresser and Anders for a qubit coupled along $\hat{\mathbf{r}}$ is

$$\rho_S^{(\lambda\to\infty)} = \frac{1}{2}\Big[\mathbf{1} - \sigma_{\hat{\mathbf{r}}}\tanh\big(\tfrac{1}{2}\beta\omega_q\cos\theta\big)\Big]. \tag{D17}$$

In our notation $\hat{\mathbf{r}} \equiv \hat{\mathbf{f}}$. The MFG-compatible branch is therefore the continuation whose endpoint lies on the *coupling-direction sector* (the θ ray in the plotting convention), not the continuation that returns to the bare ground-state ray.

This can be made explicit by comparing the exact equilibrium ultrastrong endpoint to the MFG target. From Appendix **??**,

$$\Theta(\chi) = \operatorname{arctanh}\big(\cos\eta\,\tanh\chi\big) - a, \qquad a = \beta\omega_q/2, \tag{D18}$$

so at fixed (β,θ),

$$\Theta_\infty = \lim_{g\to\infty}\Theta = \operatorname{arctanh}(\cos\eta) - a, \tag{D19}$$

because $\chi = g^2\chi_0 \to \infty$ implies $\tanh\chi \to 1$. Substituting into Eq. (D2) gives the exact equilibrium ultrastrong endpoint

$$\mathbf{m}_\infty^{(\rm ex)} = \frac{\sin\eta\,\hat{\mathbf{e}}_\perp^{(s)} + \big(\cos\eta\cosh a - \sinh a\big)\hat{\mathbf{n}}_s}{\cosh a - \cos\eta\,\sinh a}, \tag{D20}$$

with exact purity

$$r_\infty^{(\rm ex)} = \lim_{g\to\infty}\tanh\chi = 1. \tag{D21}$$

By contrast, the MFG endpoint in our notation is

$$\rho_{S,\infty}^{(\rm MFG)} = \frac{1}{2}\big[\mathbf{1} - \sigma_{\hat{\mathbf{f}}}\,r_{\rm MFG}\big], \qquad r_{\rm MFG} \equiv \tanh(a\cos\theta), \tag{D22}$$

so MFG consistency requires both

$$\varphi_{\rm ax}^{(\rm br)}(g\to\infty) = \theta \pmod{\pi}, \tag{D23}$$

$$r_{\rm br}(g\to\infty) = \tanh(a\cos\theta). \tag{D24}$$

Equation (D21) then shows the obstruction directly: a pure chart relabeling can match the *direction*, but not the finite-temperature MFG *radius*.

Correlated-limit viewpoint and branch selection. The physically persuasive way to state the MFG comparison is to keep the effective thermal variable

$$a_{\rm eff} \equiv \frac{\beta\omega_q}{2}\cos\theta \tag{D25}$$

visible in the asymptotic endpoint. Then the MFG radius is simply

$$r_{\rm MFG} = \tanh a_{\rm eff}, \tag{D26}$$

with density-matrix purity

$$\mathcal{P}_{\rm MFG} = \frac{1}{2}\big(1 + r_{\rm MFG}^2\big). \tag{D27}$$

In this correlated-limit language the endpoint radius is fixed by $a_{\rm eff}$, and the remaining freedom is angular. The two post-hub continuations therefore correspond to distinct endpoint directions:

$$\mathbf{m}_{\rm br}^{(\rm ground)} \to -\tanh(a_{\rm eff})\,\hat{\mathbf{n}}_s, \tag{D28}$$

$$\mathbf{m}_{\rm br}^{(\rm MFG)} \to -\tanh(a_{\rm eff})\,\hat{\mathbf{f}}. \tag{D29}$$

The second is the Cresser–Anders-compatible branch. In other words: once the endpoint is required to retain a finite thermal imprint through $a_{\rm eff}$, the continuation that runs back to the bare ground state is no longer the physically relevant branch for MFG matching.

A simple branch-family ansatz implementing this idea is

$$\mathbf{m}_{\rm br}(\beta) = r_{\rm MFG}(\beta)\left[\sin\alpha_{\rm br}(\beta)\,\hat{\mathbf{e}}_\perp^{(s)} - \cos\alpha_{\rm br}(\beta)\,\hat{\mathbf{n}}_s\right], \tag{D30}$$

where $\alpha_{\rm br}(\beta)$ is selected from the local branch family so that the trajectory passes through the hub and asymptotes to the θ sector. This is a branch-switched continuation rule, not an exact equilibrium trajectory.

Purity diagnostics: exact law versus branch-switched families. For the exact equilibrium qubit state,

$$r_{\rm exact}(\beta, g) = \sqrt{m_z^2 + m_\perp^2} = \tanh\chi(\beta, g). \tag{D31}$$

If one only relabels the angle chart, the purity is unchanged. If one instead reconstructs a new branch-switched state trajectory, the purity becomes an independent diagnostic:

$$r_{\rm br}(\beta) \equiv \sqrt{\left[m_z^{(\rm br)}(\beta)\right]^2 + \left[m_\perp^{(\rm br)}(\beta)\right]^2}. \tag{D32}$$

This distinction is essential for interpretation: branch multiplicity lives in the angle chart, while MFG matching imposes an additional radial requirement.

Figure: what should be read from it. Figure 10 is designed to isolate the three ingredients that were entangled in the folded Fig. 9(b): the geometric hub at $\pi - \theta$, the post-hub branch choice, and the radial law. Panel (a) shows the unfolded Bloch-circle geometry and the two post-hub continuations. Panel (b) shows the same construction on the angle chart. Panel (c) shows the MFG-guided radius and purity used in the ansatz. The figure is intentionally presented as a branch-family construction rather than an exact equilibrium trajectory, because that is precisely the point of the MFG comparison.

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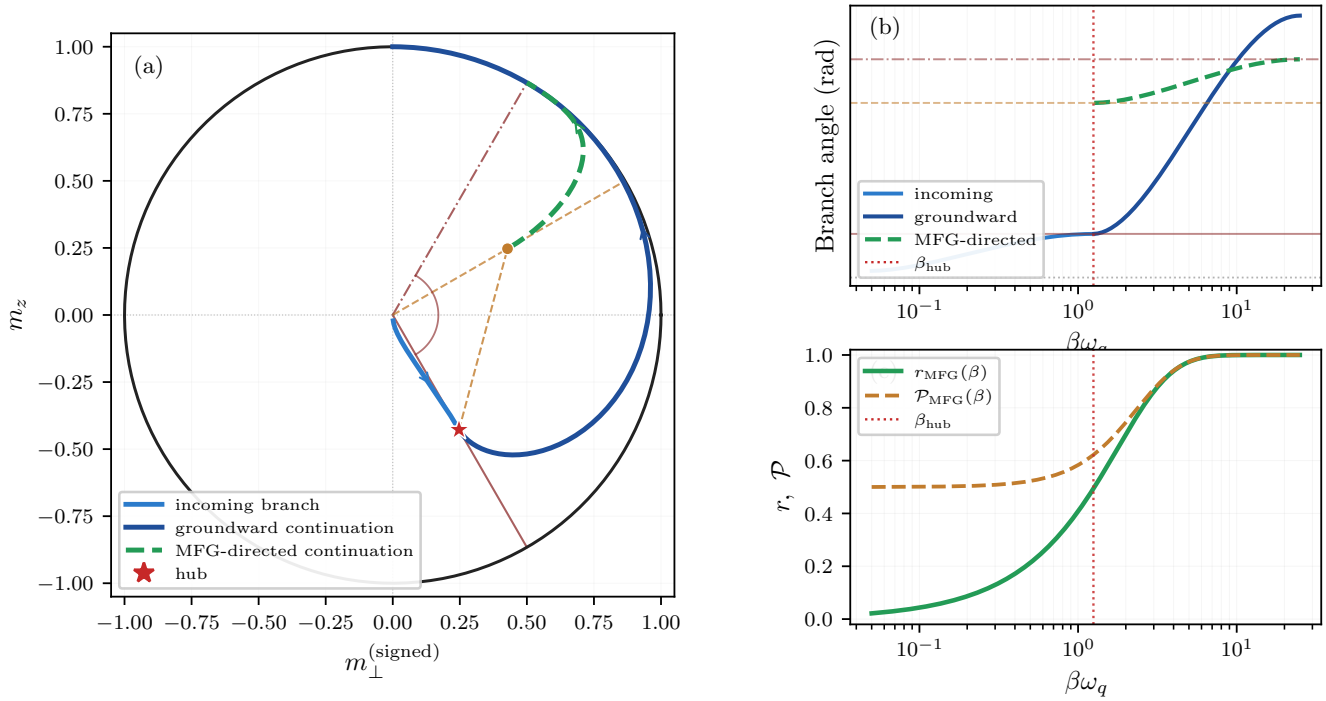

FIG. 10. Hub-centered branch family illustrating the tilt-angle interpretation and MFG-directed continuation (illustrative ansatz, not an exact equilibrium trajectory). (a) Unfolded Bloch-circle geometry in $(m_\perp^{(\mathrm{signed})}, m_z)$: an apparent branch reaches the hub $\varphi_{\mathrm{ax}} = \pi - \theta$, after which one continuation returns groundward while the alternate sign/arctan image (starting at the local 2θ point) precesses toward the coupling-direction sector θ. (b) The corresponding angle-branch construction. (c) The MFG-guided Bloch radius and density-matrix purity used in the branch-family ansatz, $r_{\mathrm{MFG}} = \tanh[(\beta\omega_q/2)\cos\theta]$ and $\mathcal{P}_{\mathrm{MFG}} = \frac{1}{2}(1 + r_{\mathrm{MFG}}^2)$.

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[57] There is a third limit, but it lies outside the present scope of our discussion.